ECE TYPE-APPROVAL CERTIFICATE

E24

Communication Concerning: Approval granted
~~Approval extended~~
~~Approval refused~~
~~Approval withdrawn~~
~~Production definitively discontinued~~

Of a type of electrical/electronic sub-assembly with regard to Regulation No.10.

Approval No: ***E24*10R06/01*3870*00***

Reason for extension: ***-N/A***

1. Make (trade name of manufacturer): inhand

2. Type and general commercial description: ***VG814***
InVehicle Gateway

Variant(s): ***N/A***

3. Means of identification of type, if marked on the ~~vehicle~~/component/~~separate technical unit~~: ***Approval mark***

3.1 Location of that marking: ***Stuck on the shells, See Drawings of the ESA***

4. Category of vehicle: ***N/A***

5. Name and address of manufacturer: ***Beijing InHand Networks Technology Co., Ltd.***
Room 501, 5th floor , building 3, No.18 Yard, Ziyue Road, Chaoyang District, Beijing, China 100102

6. In the case of components and separate technical units, location and method of affixing of the approval mark: ***Stuck on the shells, See Drawings of the ESA***

7. Address(es) of assembly plant(s): ***Inhand Jiaxing Communication Technology Co., Ltd.***
318 Ruifeng Street, Gaozhao Street, Xiuzhou District, Jiaxing City, Zhejiang Province, China

Approval No: ***E24*10R06/01*3870*00***

8. Additional information (where applicable): ***See appendix below***

9. Technical service responsible for carrying out the tests: ***CETOC Technical Service srl***
Via della Bufalotta, 374
00139 Roma

10. Date of test report: ***10.12.2021***

11. Number of test report: ***CN-50-17-151-COM22-02943-FIR***

12. Remarks (if any): ***See Appendix below***

13. Place: ***Dublin***

14. Date: ***30th March, 2022***

15. Signature:

16. The index to the information package lodged with the approval authority, which may be obtained on Request, is attached.

Approval No: ***<u>E24*10R06/01*3870*00</u>***

Appendix

To type-approval communication concerning the type approval
of an electrical/electronic sub-assembly under Regulation No.10.

1.	Additional information	
1.1.	Electrical system rated voltage:	***DC 12/24V, Negative ground***
1.2.	This ESA can be used on any vehicle type with the following restrictions:	***See manufacturer's specifications.***
1.2.1	Installation conditions, if any:	***See manufacturer's specifications.***
1.3.	This ESA can only be used on the following vehicle types:	***N/A***
1.3.1	Installation conditions, if any:	***N/A***
1.4.	The specific test method(s) used and the frequency ranges covered to determine immunity were:	***Bulk Current Injection Method:*** *Frequency: (20 – 400 MHz)* ***Absorber Chamber Test:*** *Frequency: (400 – 2000 MHz)*
1.5.	Laboratory accredited to ISO 17025 and recognized by the Approval Authority responsible for carrying out the tests:	***CETOC Technical Service srl***
2.	Remarks:	***N/A***

Appendix to type-approval communication concerning the
type approval of a vehicle under Regulation No.10.

1.	Additional information	
2.	Special devices for the purpose of Annex 4 to this Regulation:	***N/A***
3.	Electrical system rated voltage:	***N/A***
4.	Type of bodywork:	***N/A***
5.	List of electronic systems installed in the tested vehicle(s) not limited to the items in the information document:	***N/A***
5.1	Vehicle equipped with 24 GHz short-range radar equipment (yes/no):	***N/A***
6.	Laboratory accredited to ISO 17025 and recognized by the Approval Authority responsible for carrying out the tests:	***N/A***
7.	Remarks:	***N/A***

Approval No: ***E24*10R06/01*3870*00***

Index to the Information Package

Date of issue:	***30th March, 2022***
Date of latest amendment:	***N/A***
Reason for extension/revision:	***N/A***

1. Additional conditions, and advisory notes on legal alternatives.

2. Test report(s)

- numbers(s):	***CN-50-17-151-COM22-02943-FIR***
- date of issue:	***01.03.2022***
- date of latest amendment:	***N/A***

3. Information document

- number(s):	***VG814-00***
- date of issue:	***24.02.2022***
- date of latest amendment:	***N/A***
Documentation:	***45 pages***

NSAI, 1 Swift Square, Northwood, Santry, Dublin 9, Ireland. Telephone: (+353+1) 807 3800, Facsimile: 01-807 3844

Approval No: ***E24*10R06/01*3870*00***

Appendix: **Additional conditions, and advisory notes on legal alternatives**

A: Additional conditions:

1. The attached technical report, with any of its attachments, forms part of this Type Approval certificate.
2. Each device from series production shall be to the measurements specified in the attached drawings, and shall be manufactured only from the materials specified in the Approval documents.
3. Changes in the type are permitted only with the explicit permission of NSAI. Breaches of this requirement will lead to a withdrawal of the Type Approval, and in addition may be subject to criminal prosecution.
4. At regular intervals, any tests or associated checks prescribed by the applicable legislation to verify continued conformity with the approved type shall be carried out. The manufacturer shall demonstrate compliance with this by submitting to NSAI evidence of adequate arrangements and documented control plans for each type approved.
5. Any set of samples or test pieces showing evidence of non-conformity shall give rise to further sampling and testing and all steps shall be taken to restore conformity of production.
6. This Type Approval will expire when it is surrendered by the holder, or withdrawn by NSAI, or when the approved type no longer conforms to legal requirements. The recall of the Type Approval can be issued by NSAI when the conditions required for the issuing or continuation of the Type Approval are no longer current, or when the Approval holder is in breach of the duties attached to the Type Approval, or when it is established that the approved type no longer meets the requirements of traffic safety.
7. Changes in the company name, address or manufacturing site, as well as in any of the sales or other agents specified in the issuing of the approval must immediately be notified to NSAI.
8. The duties imposed by the issuing of this certificate are not transferable. The legal protection of third parties is not affected by this certificate.
9. When the manufacture or sale of the system, component or separate technical unit has not been started within one year of the date of issue of this certificate, then NSAI is to be informed. This requirement also applies when the manufacture or sale has been halted for more than one year, or when it ought to have been halted for more than one year. The initial commencement of manufacture or sale, or the resumption of manufacture or sale, shall then be notified to NSAI within one month of commencement or resumption.

B: Legal Options:

Any objection to the requirements set out in this certificate shall be made within one month of the date of issue. The objection shall be made, in writing, to NSAI in Dublin.

NSAI, 1 Swift Square, Northwood, Santry, Dublin 9, Ireland. Telephone: (+353+1) 807 3800, Facsimile: 01-807 3844

Electromagnetic Compatibility – ESA

0.	**Legislation:**		
0.1.	Requirements according to	:	UNECE Regulation 10.06 to Supplement 1
1.	**General**		
1.1.	Reason for Inspection Report	:	New approval / ~~Extension of approval / Test report only / COP~~
1.2.	Manufacturer's Representative(s)	:	No attendance
1.3.	CETOC TS Representative(s)	:	Lily Li
1.4.	Location of Test	:	GuangZhou ShunTai Quality Technical Service Co., Ltd. Room 206A, 2F, 4# Plant, No.63 Punan Road, Huangpu District, Guangzhou, China, 510760
1.5.	Data of test	:	27/02/2022
2.	**Manufacturer Details**		
2.1.	Make	:	inhand
2.2.	Type	:	VG814
2.3.	Variant/Version	:	N/A
2.4.	Commercial Name	:	InVehicle Gateway
2.5.	Category	:	Component
2.6.	Name and Address of manufacturer	:	Beijing InHand Networks Technology Co., Ltd. Room 501, 5th floor , building 3, No.18 Yard, Ziyue Road, Chaoyang District, Beijing, China 100102
3.	**Conclusion:**		
3.1.	Final conclusion of the inspection:		The above mentioned type was tested in accordance with the above mentioned legislation and was found to comply in all respects. This Inspection report relates only to the items tested.

Signature:	:	Lily Li	
Name:	:	Lily Li	Massimo Peraboni
Position:	:	Type Approval Engineer	Tech. Mgr.
Place and date:	:	Guangzhou, 01/03/2022	Roma, 01/03/2022

4. List of annexes:

APPENDIX 1 - TEST REPORT HISTORY

List this report and previous reports, with extension details.

Inspection Report Number	Reason for Extension	Date of Issue
CN-50-17-151-COM22-02943-FIR	N/A	01/03/2022

APPENDIX 2 – GENERAL SPECIFICATION

1.	**Worst Case Rationale**	:	All tests have been performed with both rated voltages: DC 12V/24V.
2.	**Significant Interpretations, Alternative Test Methods, New Technologies**	:	N/A
3.	**Summary of test results**		
3.1.	Applicability	:	

	PASS	FAIL	N/A
Radiated Emissions:	☒	☐	☐
Radiated Immunity:	☒	☐	☐
BCI Immunity:	☒	☐	☐
Free Field Immunity:	☒	☐	☐
150 mm Stripline Immunity:	☐	☐	☒
800 mm Stripline Immunity:	☐	☐	☒
Transient Testing:	☒	☐	☐

4. **Component Specification**

Component Part Number:	VG814

5.	**Facility and Equipment Checks**		
5.1.	Calibration certificates checked and valid, recorded in the following table	:	Conform
5.2	All instruments are equipped with identification label	:	Yes
5.3	Calibration certificates are complete of calibration-chain with detailed information regarding primary used.	:	Yes
5.4.	Is the anechoic/semi-anechoic chamber correctly set up in all its electrical/mechanical parts to ensure the validity of the measurements?	:	Yes

Equipment	Serial / Certificate No.	Calibration due*
ALSE ROOM	CN 2GB20060423919-0001	05/06/2022
EMI receiver	CN 2GB21071389810-0040	23/07/2022
L.I.S.N.	CN 2GB21071389810-0042	23/07/2022
L.I.S.N.	CN 2GB21071389810-0043	23/07/2022
Biconical antenna	CN J202108232679-0007	12/09/2022
Log-periodic antenna	CN J202108232679-0008	12/09/2022
Supply Voltage Change Simulator	CN 2GB21071389810-0026	23/07/2022
Load dump wave simulator	CN 2GB21071389810-0027	23/07/2022
Transient pulse disturbance simulator	CN 2GB21071389810-0028	23/07/2022
Current Injection Clamp	CN 2GB21071389810-0047	23/07/2022
Digital phosphor oscilloscope	CN 2GB21071389810-0033	23/07/2022

**Specify calibrated date + (interval) or calibration due date.*

CETOC Technical Service srl
Via della Bufalotta, 374,
00139 Roma

APPENDIX 3 – INSPECTION RESULTS

		PASS	FAIL	N/A
	Radiated Emissions			
CISPR25, 4.5.	Measuring equipment complies with CISPR 16-1-4 (2010).	☒	☐	☐
	Test Location			
Ann 7, 3.1. *Ann 7, 3.3.*	Test performed in: - A.L.S.E (Absorber-lined Shielded Enclosure)* - ~~O.A.T.S (Open Area Test Site)*~~ **Strikethrough, as appropriate.*	☒	☐	☐
Ann 7, 3.3.	O.A.T.S level is a clear area, free from electromagnetic reflecting surfaces, within a circle of 15 m minimum radius.	☐	☐	☒
Ann 7, 3.3.	Measuring equipment is outside 15 m minimum radius circle.	☐	☐	☒
Ann 7, 3.4.	Ambient noise is at least 6 dB below reference limits, in either case.	☒	☐	☐
	Test Arrangements			
CISPR25, 4.4.2.	EUT and antenna are more than 2 m from the walls and ceiling, and 1 m from the nearest absorber material.	☒	☐	☐
CISPR25, 6.1.1.	Ground plane is 900 ± 50 mm high and made from 0.5 mm thick copper, brass or galvanised steel.	☒	☐	☐
CISPR25, 6.1.1.	Ground plane is at least 2,000 mm length x 1,000 mm width.	☒	☐	☐
CISPR25, 6.4.2.3.	ESA and harness are supported at 50 ± 5 mm above the ground plane on low relative permittivity material.	☒	☐	☐
CISPR25, 6.4.2.3.	Face of the ESA is within 200 mm ± 10 mm from the edge of the ground plane.	☒	☐	☐
CISPR25, 6.4.2.4.	Length of test harness, parallel to the front of the ground plane, is 1,500 ± 75 mm and does not exceed 2,000 mm.	☒	☐	☐
CISPR25, 6.4.2.4.	Long segment of test harness is located parallel to the edge of the ground plane, facing the antenna at a distance of 100 ± 10 mm from the edge.	☒	☐	☐
CISPR25, 6.1.2.	Power supply is Artificial Network (AN) rated at 5 Ω/50 μH.	☒	☐	☐
CISPR25, 6.1.2.	EUT is: - Remotely grounded (vehicle power return line longer than 200 mm): two artificial networks are required, one for the positive supply line and one for the power return line* ~~- Locally grounded (vehicle power return line 200 mm or shorter): one artificial network is required for the positive supply*~~ **Strikethrough, as appropriate.*			
CISPR25, 6.1.2.	Case of the ESA is: ~~- Grounded, simulating actual vehicle configuration*~~ - Not grounded, simulating actual vehicle configuration* **Strikethrough, as appropriate.*	☒	☐	☐
CISPR25, 6.1.2.	AN is electrically bonded to the ground plane.	☒	☐	☐

	Antenna			
CISPR25, 6.4.2.6.	Height of the phase centre is 100 ± 10 mm above the ground plane.	☒	☐	☐
CISPR25, 6.4.2.6.	No part of any antenna radiating element is closer than 250 mm to the floor.	☒	☐	☐
CISPR25, 6.4.2.6.	Radiating elements of the measuring antenna are not closer than 1,000 mm to any absorber material, except that used on the floor, and are not closer than 2,000 mm to the walls or ceiling of the shielded enclosure.	☒	☐	☐
CISPR25, 6.4.2.6.	Phase centre (for biconical) or tip (for log-periodic) is 1,000 ± 50 mm from the harness.	☒	☐	☐
CISPR25, 6.4.2.6.	Antenna calibrated for this distance to correct measuring point (phase centre or tip).	☒	☐	☐
CISPR25, 6.4.2.6.	Phase centre of the antenna is in line with the centre of the longitudinal part of the wiring harness.	☒	☐	☐
Ann 7, Ann 8, 4.3.	Pre-test sweep supplied to show compliance throughout frequency range 30 to 1,000 MHz.	☒	☐	☐
Ann 7, Ann 8, 4.3.	Test frequencies chosen from pre-test data.	☒	☐	☐
	Narrowband Test Results			
Ann 8, 2.	Operational mode of ESA: Normal operation	☒	☐	☐
Ann 8, 4.2.	Detector used and bandwidth: Average, 120kHz	☒	☐	☐
6.6.2.	ESA meets narrowband emissions limits, with both vertical and horizontal polarisations.	☒	☐	☐
	Broadband Test Results			
Ann 7, 2.	Operational mode of ESA: Normal operation	☒	☐	☐
Ann 7, 4.2.	Detector used and bandwidth: Quasi peak, 120kHz	☒	☐	☐
6.5.2.	ESA meets broadband emissions limits, with both vertical and horizontal polarisations.	☒	☐	☐
	Radiated Immunity			
	Test Method(s) used and Frequency Range(s)			
ISO11452-4	BCI frequency range between 20 and 400 MHz:	☒	☐	☐
ISO11452-2	Free field frequency range between 400 and 2,000 MHz:	☒	☐	☐
ISO11452-3	TEM cell frequency range between 20 and 200 MHz:	☐	☐	☒
ISO11452-5	150 mm stripline frequency range between 20 and 400 MHz:	☐	☐	☒
ISO11452-5	800 mm stripline frequency range between 20 and 2,000 MHz:	☐	☐	☒

Maximum frequency step sizes do not exceed:

Frequency Band (MHz)	Linear Steps (MHz)	Log Steps (%)	Actual Steps Used
20 - 200	5	5	5%
200 - 400	10	5	5%
400 - 1000	20	2	2%
1000 - 2000	40	2	2%

Test Arrangements (General)

Ann 9, 2.2.	Operational mode of ESA: Normal operation	☒	☐	☐
Ann 9, 2.3.	Extraneous equipment in place during calibration.	☒	☐	☐
Ann 9, 2.4.	Test equipment used is the same as for calibration.	☒	☐	☐
Ann 9, 2.5.	Loads and actuators are as realistic as possible.	☒	☐	☐
Ann 9, 2.5.	Case of ESA is: - ~~Grounded, simulating actual vehicle configuration*~~ - Not grounded, simulating actual vehicle configuration* **Strikethrough, as appropriate.*	☒	☐	☐
Ann 9, 3.1.	Test frequency range is 20 to 2,000 MHz.	☒	☐	☐
Ann 9, 3.1.	Test signal is R.F. sine wave amplitude, modulated by a 1 kHz sine wave at a modulation depth of 0.8 ± 0.04, in the 20 - 800 MHz band and pulse modulation (time on 577 µs, period 4,600 µs) in the 800 – 2,000 MHz band.	☒	☐	☐
6.8.2.1.	Pre-test sweep supplied to show compliance throughout frequency range 20 to 2,000 MHz.	☒	☐	☐
Ann 9, 3.2.	Test frequencies chosen from pre-test data.	☒	☐	☐
6.8.2.2.	No degradation of immunity related functions during the tests.	☒	☐	☐

BCI Immunity

ISO11452-4, 5.	Shielded area used: Yes	☒	☐	☐
ISO11452-4, 8.3.2.1.	Forward power used to achieve specified current.	☒	☐	☐

Installation of ESA under Test

Ann 9, 4.3.2.	Current probe located 150 ± 10 mm from ESA connectors.	☒	☐	☐
Ann 9, 4.3.2.	ESA installed: - ~~In a vehicle, as per ISO 11451-4*~~ - On a ground plane, as per ISO 11452-4* **Strikethrough, as appropriate.*	☒	☐	☐
ISO11452-4, 7.1.	Ground plane is made from at least 0.5 mm thick copper, brass or galvanised steel.	☒	☐	☐
ISO11452-4, 7.1.	Minimum width of the ground plane is 1,000 mm and the minimum length is 1,500 mm, or length of the entire underneath of equipment plus 200 mm, whichever is greater.	☒	☐	☐

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ISO11452-4, 7.1.	Height of the ground plane is 900 ± 100 mm.	☒	☐	☐
ISO11452-4, 7.1.	Ground plane is bonded to the shielded enclosure, with the straps at a distance no greater than 300 mm apart.	☒	☐	☐
ISO11452-4, 7.2.	- ESA remotely grounded (vehicle power return line longer than 200 mm): two artificial networks are required, one for the positive supply line and one for the power return line)* - ~~ESA locally grounded (vehicle power return line 200 mm or shorter): one artificial network is required, for the positive supply~~* **Strikethrough, as appropriate.*			
ISO11452-4, 7.2.	Power supply is Artificial Network (AN) rated at 50 Ω/5 µH.	☒	☐	☐
ISO11452-4, 7.3.	ESA and harness supported 50 ± 5 mm above ground plane, on low relative permittivity material.	☒	☐	☐
ISO11452-4, 7.3.	Face of the ESA within 100 mm from the edge of the ground plane.	☒	☐	☐
ISO11452-4, 7.3.	Distance of at least 500 mm between ESA and any metal parts, such as the walls of the shielded enclosure (exception is ground plane).	☒	☐	☐
ISO11452-4, 7.4.	Length of test harness is 1,000 ± 100 mm, unless specified.	☒	☐	☐
	BCI Test Results			
6.8.2.1.	No malfunction at 60 mA or below.	☒	☐	☐
	Free Field Immunity			
ISO11452-2, 8.3.1.	Test field defined by: - Forward power* - ~~Another parameter, directly related~~* **Strikethrough, as appropriate.*	☒	☐	☐
ISO11452-2, 8.3.2.	Antenna is at a distance of 1,000 ± 10 mm from the reference point.	☒	☐	☐
ISO11452-2, 8.3.2.	Reference point is 150 ± 10 mm above the ground plane.	☒	☐	☐
ISO11452-2, 8.3.2.	Reference point is 100 ± 10mm from the edge of the ground plane.	☒	☐	☐
ISO11452-2, 8.3.2.	For frequencies from 80 - 1,000 MHz, the reference point is in the centre of the harness.	☒	☐	☐
ISO11452-2, 8.3.2.	For frequencies from 1,000 - 2,000 MHz, the reference point is in line with the ESA.	☒	☐	☐
	Test Arrangements			
ISO11452-2, 7.1.	Ground plane is made from at least 0.5 mm thick copper, brass or galvanised steel.	☒	☐	☐
ISO11452-2, 7.1.	Minimum width of the ground plane is 1,000 mm and the minimum length is 2,000 mm.	☒	☐	☐
ISO11452-2, 7.1.	Height of the ground plane is 900 ± 100 mm.	☒	☐	☐

ISO11452-2, 7.1.	Bonding straps are at a distance no greater than 300 mm apart.	☒	☐	☐
ISO11452-2, 7.2.	Power supply is Artificial Network (AN) rated at 50 Ω/5 μH.	☒	☐	☐
ISO11452-2, 7.2.	- ESA remotely grounded (vehicle power return line longer than 200 mm): two artificial networks are required, one for the positive supply line and one for the power return line)* - ~~ESA locally grounded (vehicle power return line 200 mm or shorter): one artificial network is required, for the positive supply~~* **Strikethrough, as appropriate.*			
ISO11452-2, 7.3.	AN mounted directly on the ground plane and cases bonded to the ground plane.	☒	☐	☐
ISO11452-2, 7.3.	ESA and harness supported 50 ± 5 mm above table, on low relative permittivity material.	☒	☐	☐
ISO11452-2, 7.3.	Face of the ESA located 200 ± 10 mm from the edge of the ground plane.	☒	☐	☐
ISO11452-2, 7.4.	Test harness parallel to the front edge of the ground plane.	☒	☐	☐
ISO11452-2, 7.4.	Total length of harness does not exceed 2,000 mm.	☒	☐	☐
ISO11452-2, 7.4.	Actual wiring harness length: N/A m	☐	☐	☒
	<u>or</u> Length is 1,500 ± 75 mm between ECU and AN.	☒	☐	☐
ISO11452-2, 7.4.	Harness is at a distance of 100 ± 10 mm from the edge of the ground plane.	☒	☐	☐
ISO11452-2, Fig 1	Front face of ESA is at least 1.0 m from all other conductive structures.	☒	☐	☐
ISO11452-2, Fig 1	ESA harness is at least 2.0 m forward from the chamber wall.	☒	☐	☐
	Antenna Type(s) and Frequency Range(s)			
Ann 9, 4.1.2.	Antenna is vertically polarised.	☒	☐	☐
ISO11452-2, 7.6.	Antenna is in the same position as the calibration.	☒	☐	☐
ISO11452-2, 7.6.	Phase centre is 100 ± 10 mm above the ground plane.	☒	☐	☐
ISO11452-2, 7.6.	Antenna elements are no closer than 250 mm to the floor of the facility, no closer than 0.5 m to any radio absorbent material, and no closer than 1.5 m to the wall of the facility.	☒	☐	☐
ISO11452-2, 7.6.	Distance between wiring harness and antenna is 1,000 mm ± 10 mm, measured from the phase-centre of the biconical antenna, or the nearest part of the log-periodic and horn antennas.	☒	☐	☐
Ann 9, 3.1.	Test signal modulation is: - AM, 1 kHz modulation, 80 % depth in 20 - 800 MHz frequency range; - PM, ton 577 μs, period 4,600 μs in 800 - 2,000 MHz frequency range.	☒	☐	☐
	Free Field Immunity Test Results			
6.8.2.	No malfunction at 25 V/m or below.	☒	☐	☐
	150 mm Stripline Immunity			
ISO11452-5, 5.3.1.	Stripline housed in a shielded room.	☐	☐	☒

ISO11452-5, 6.2.2.	Test field defined by: - Forward power* - Another parameter, directly related* **Strikethrough, as appropriate.*	☐	☐	☒
ISO11452-5, 6.2.3.	Field probe in the centre of stripline.			
	Installation of ESA under Test			
ISO11452-5, 5.3.1.	ESA is 200 + 20 - 0 mm from the edge of the active conductor.	☐	☐	☒
ISO11452-5, 5.3.1.	Peripherals are a minimum of 200 mm from the edge of the active conductor.	☐	☐	☒
ISO11452-5, 5.3.1.	Harness supported 50 mm above the ground plane and is placed in the centre of the stripline.	☐	☐	☒
ISO11452-5, 5.3.1.	Actual wiring harness length: N/A m	☐	☐	☒
	or Minimum length under stripline is 1,000 mm.	☐	☐	☒
ISO11452-5, 5.3.1.	All wires in the harness are terminated or open, according to the vehicle application.	☐	☐	☒
ISO11452-5, 5.3.1.	Device and peripherals connected to the ground plane, as specified by the vehicle installation.	☐	☐	☒
ISO11452-5, 5.3.1.	Power supply is Artificial Network (AN) rated at 50 Ω/5 μH.	☐	☐	☒
ISO11452-5, 5.3.1.	- ESA remotely grounded (vehicle power return line longer than 200 mm): two artificial networks are required, one for the positive supply line and one for the power return line)* - ~~ESA locally grounded (vehicle power return line 200 mm or shorter): one artificial network is required, for the positive supply*~~ **Strikethrough, as appropriate.*			
	150 mm Stripline Test Results			
6.8.2.	No malfunction at 50 V/m or below.	☐	☐	☒
	800 mm Stripline Immunity			
Ann 9, 4.5.2.1.	Stripline housed in a screened room.	☐	☐	☒
Ann 9, 4.5.2.1.	Stripline positioned a minimum of 2,000 mm from the walls or metallic enclosure.	☐	☐	☒
Ann 9, 4.5.2.1.	Stripline placed on non-conducting supports at least 400 mm above the floor.	☐	☐	☒
Ann 9, 4.5.2.2.	Field probe positioned within the central one-third of the longitudinal, vertical and transverse dimensions of the space between the parallel plates, with the system under test absent.	☐	☐	☒
Ann 9, 4.5.2.2.	Test field defined by: - Forward power* - Another parameter, directly related* **Strikethrough, as appropriate.*	☐	☐	☒

Installation of ESA under Test

Ann 9, 4.5.2.3.	ESA is within the central one-third of the stripline.	☐	☐	☒
Ann 9, 4.5.2.3.	ESA is supported on non-conducting material.	☐	☐	☒
Ann 9, 4.5.2.4.	Wiring loom is arranged as per Appendix 1, Figure 3.	☐	☐	☒
Ann 9, 4.5.2.4.	Associated equipment is a minimum of 1,000 mm from stripline.	☐	☐	☒

800 mm Stripline Test Results

Frequency Suggested (MHz)	Frequency (MHz)	Forward Power		Output Level		Field Strength (V/m)
		Cal. (w)	Test (w)	Cal. (dBm)	Test (dBm)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A	N/A	N/A

6.8.2.	No malfunction at 12.5 V/m or below.	☐	☐	☒

Transient Testing

	Case of ESA is: - ~~Grounded, simulating actual vehicle configuration*~~ - Not grounded, simulating actual vehicle configuration* **Strikethrough, as appropriate.*	☒	☐	☐

Transient Immunity

6.9.1.	Test set up according to ISO 7637-2 (second edition 2004 and Amd.1:2008).	☒	☐	☐
Ann 10, 2.	Supply lines and other lines, which may be connected to supply lines, are tested.	☒	☐	☐
	Test voltage and time parameters are within allowed envelopes.	☒	☐	☐
	Test pulses and duration according to the following:	☒	☐	☐

Test Pulse	Immunity Test Level	Functional Status for Systems		Test Duration
		Related to Immunity-related Functions	Not Related to Immunity-related Functions	
1	III	C	D	5000 pulses
2a	III	B	D	5000 pulses
2b	III	C	D	10 pulses
3a	III	A	D	1 hour
3b	III	A	D	1 hour
4	III	B *(for ESA, which must be operational during engine start, or C, for other ESA)*	D	1 pulse

	ESA operational after the tests, according to the above classification.	☒	☐	☐

Emission of Conducted Disturbances

6.9.1.	Test set up according to ISO 7637-2.	☒	☐	☐
Ann 10, 3.	Supply lines and other lines, which may be connected to supply lines, are tested.	☒	☐	☐
	Slow pulses and fast pulses tested on both powering up and powering down.	☒	☐	☐

Polarity of Pulse Amplitude	Maximum Allowed Pulse Amplitude	
	Vehicles with 12 V systems	Vehicles with 24 V system
Positive	+ 75 V	+ 150 V
Negative	- 100 V	- 450 V

Remarks

None

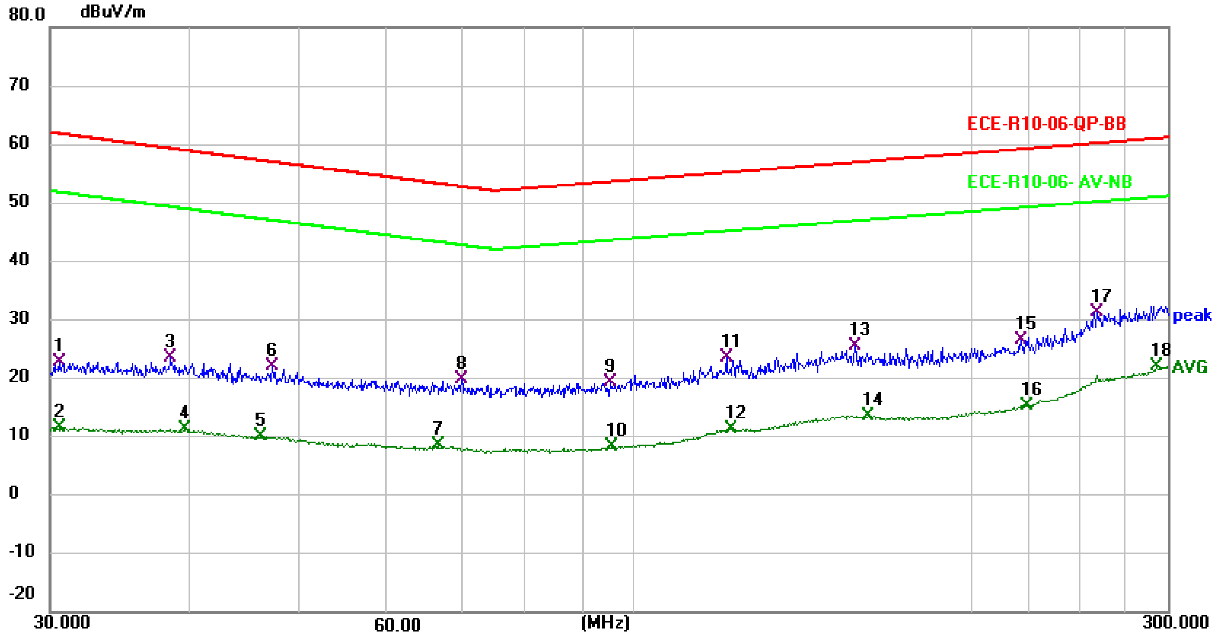
Note: CETOC TS apply measurement uncertainty to calibrated items but not test results.

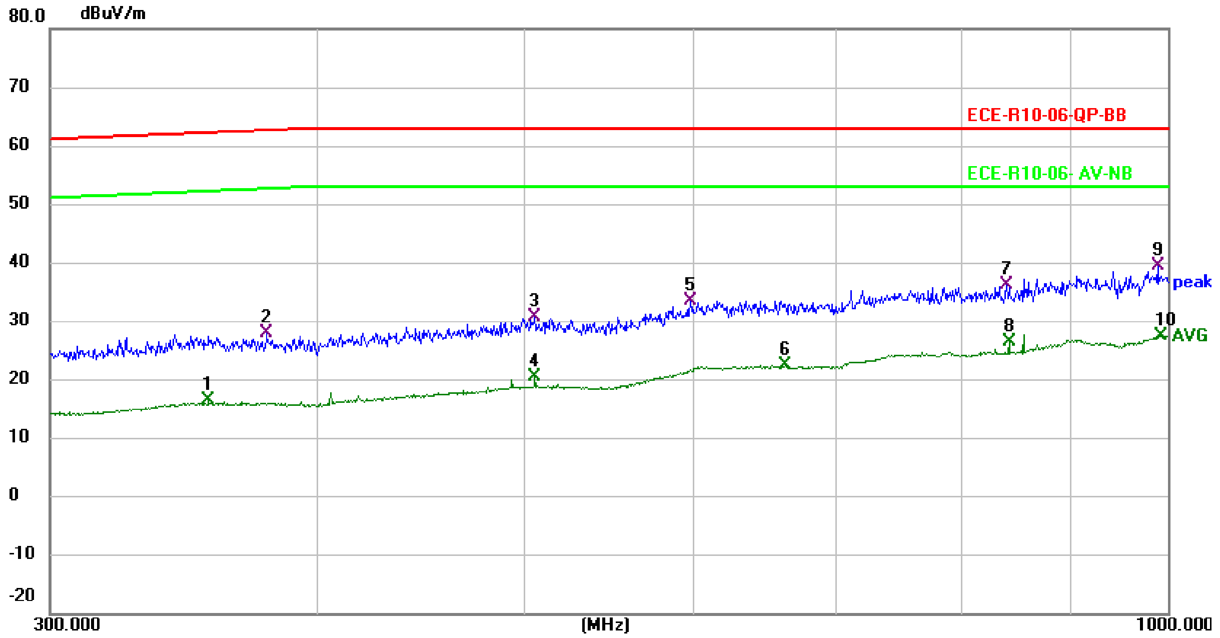
APPENDIX 4 – TEST RESULTS

APPENDIX 4.1 Radiated Emissions

Vehicles with 12V systems

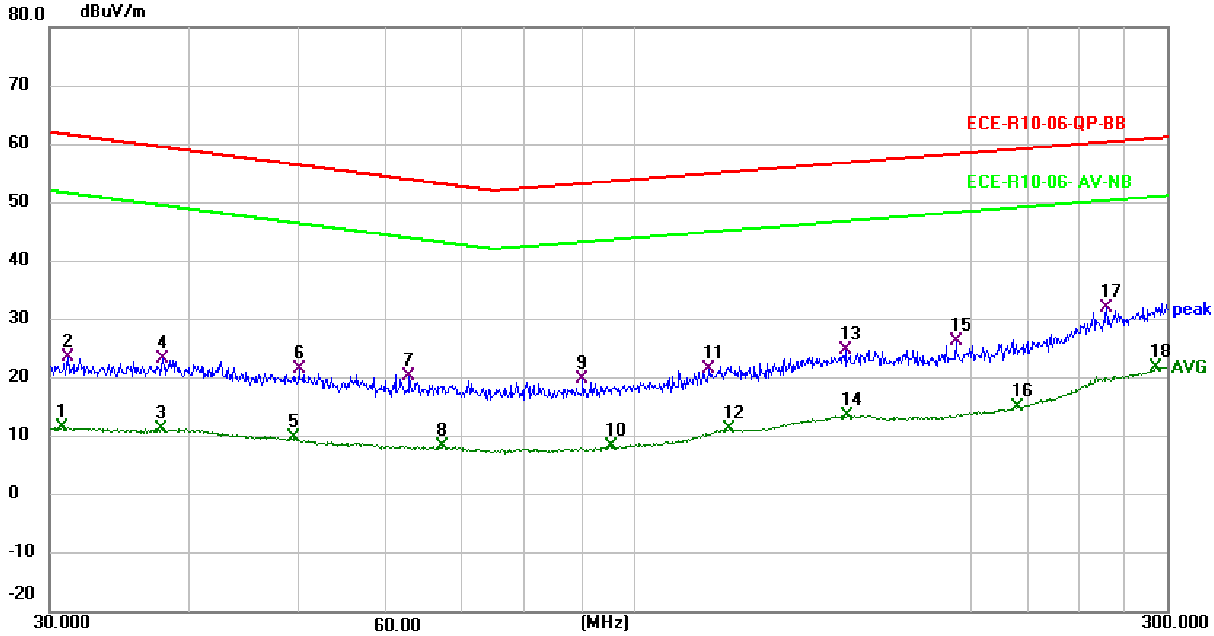
Horizontal Polarisation
30MHz to 1GHz

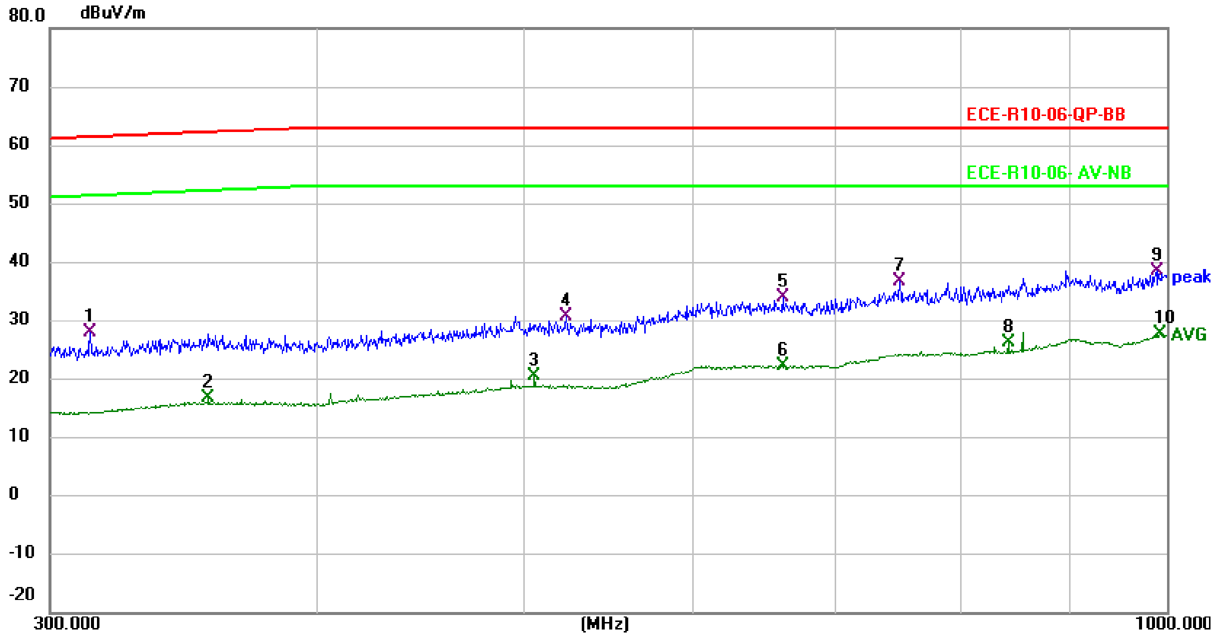




E24*10R06/01*3870*00

Vertical Polarisation
30MHz to 1GHz

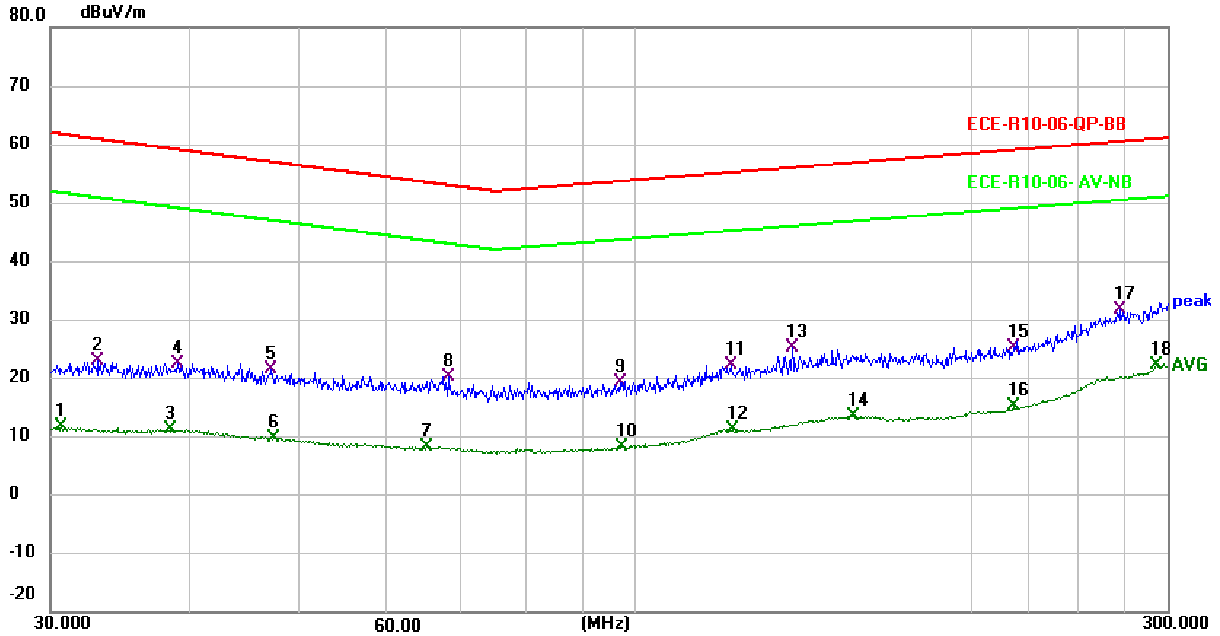


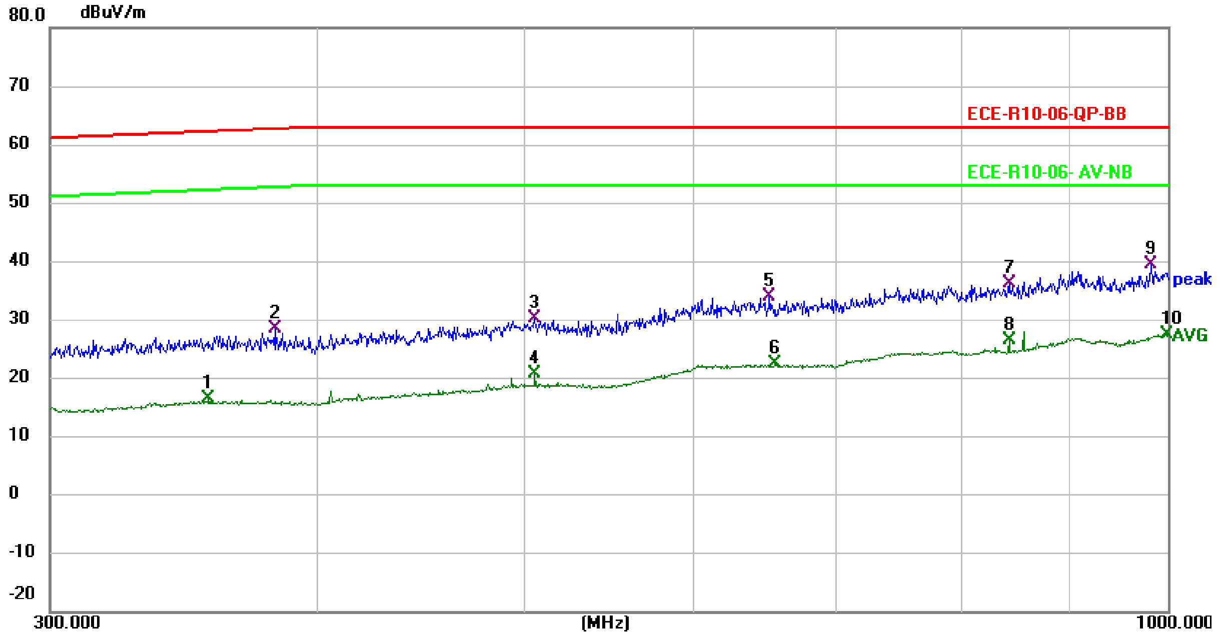


CETOC Technical Service srl
Via della Bufalotta, 374,
00139 Roma

Vehicles with 24V systems

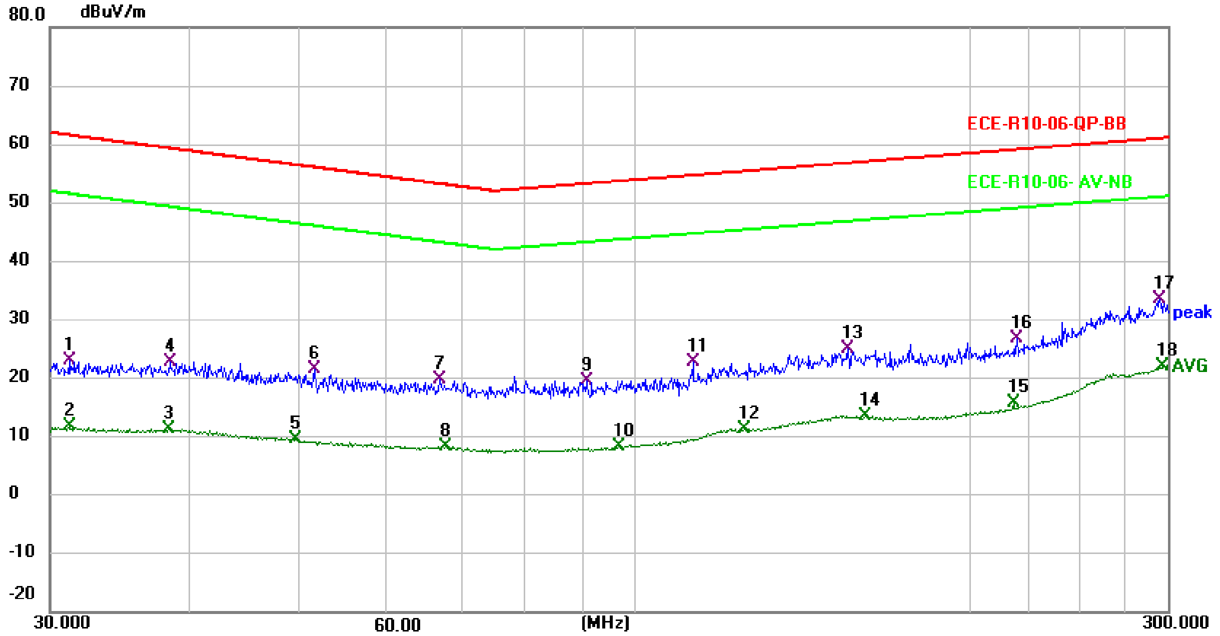
Horizontal Polarisation
30MHz to 1GHz

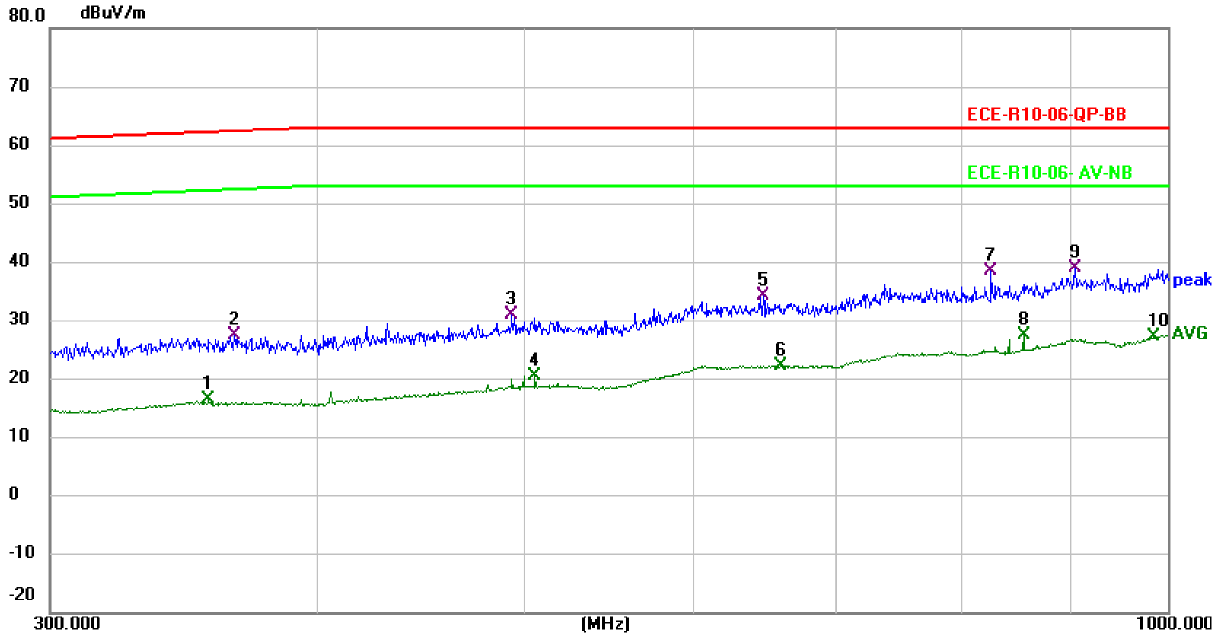




E24*10R06/01*3870*00

Vertical Polarisation
30MHz to 1GHz





APPENDIX 4.2 Radiated Immunity

Frequency (MHz)	Level (V/m)	Modulation	Polarity	Accept Status	Test Result	
					Vehicles with 12V systems	Vehicles with 24V systems
400-800	30	AM (1kHz,80%)	V	I	A	A
800-1000	30	PM	V	I	A	A
1000-2000	30	PM	V	I	A	A

APPENDIX 4.3 BCI Immunity

Frequency (MHz)	Level (mA)	Modulation	Injection place	Test Result	
				Vehicles with 12V systems	Vehicles with 24V systems
20-400	60	AM (1kHz,80%)	150mm	A	A

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00139 Roma

APPENDIX 4.4 Transient Immunity

Test Pulse	Immunity Test Level	Functional Status for Systems		Test results	
		Related to Immunity-related Functions	~~Not Related to Immunity-related Functions~~	Vehicles with 12V systems	Vehicles with 24V systems
1	III	C	~~D~~	C	C
2a	III	B	~~D~~	A	A
2b	III	C	~~D~~	C	C
3a	III	A	~~D~~	A	A
3b	III	A	~~D~~	A	A
4	III	C	~~D~~	B	B

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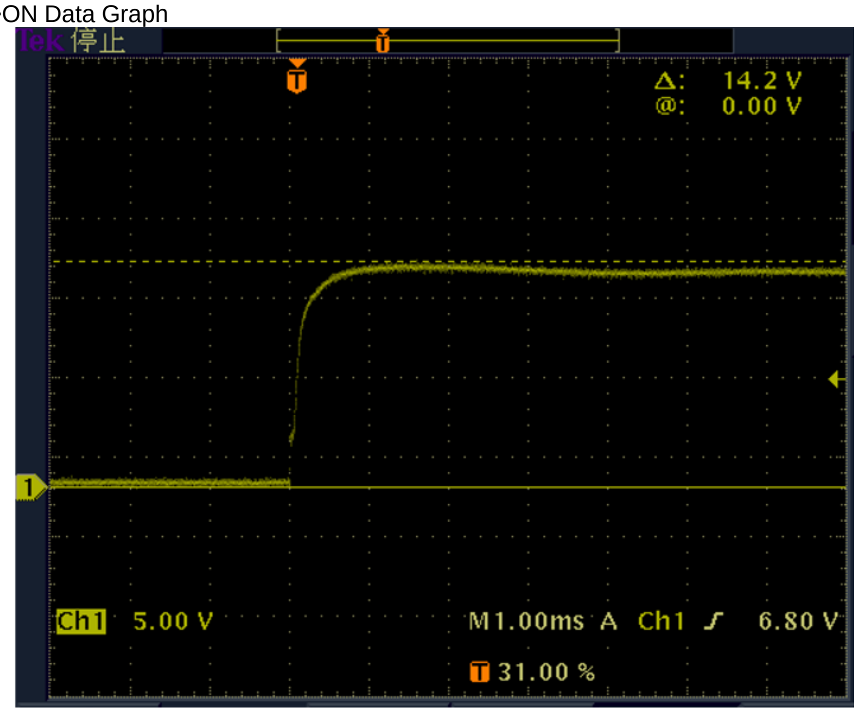
00139 Roma

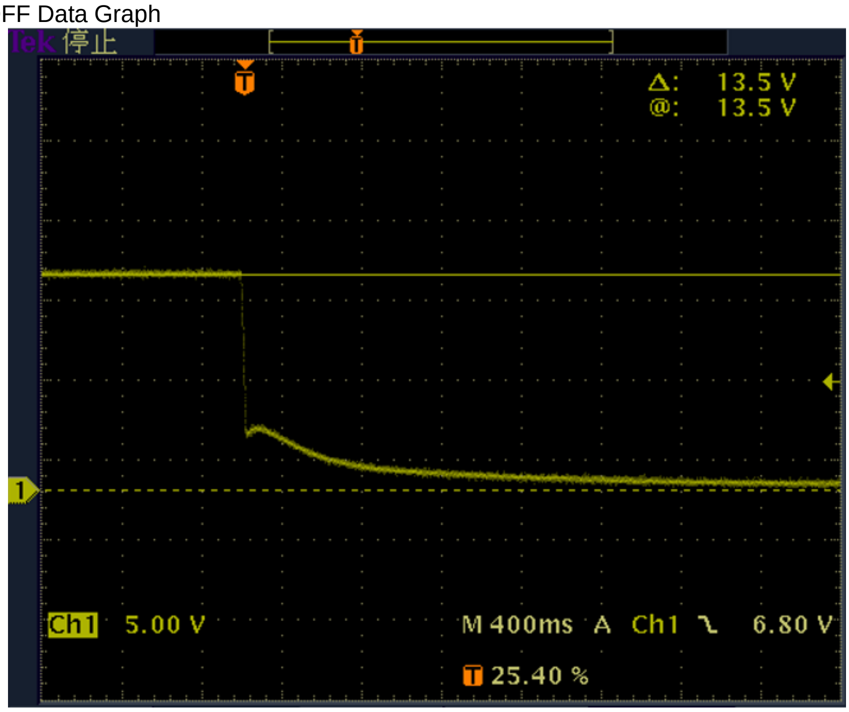
APPENDIX 4.5 Emission of Conducted Disturbances

Vehicles with 12 V systems

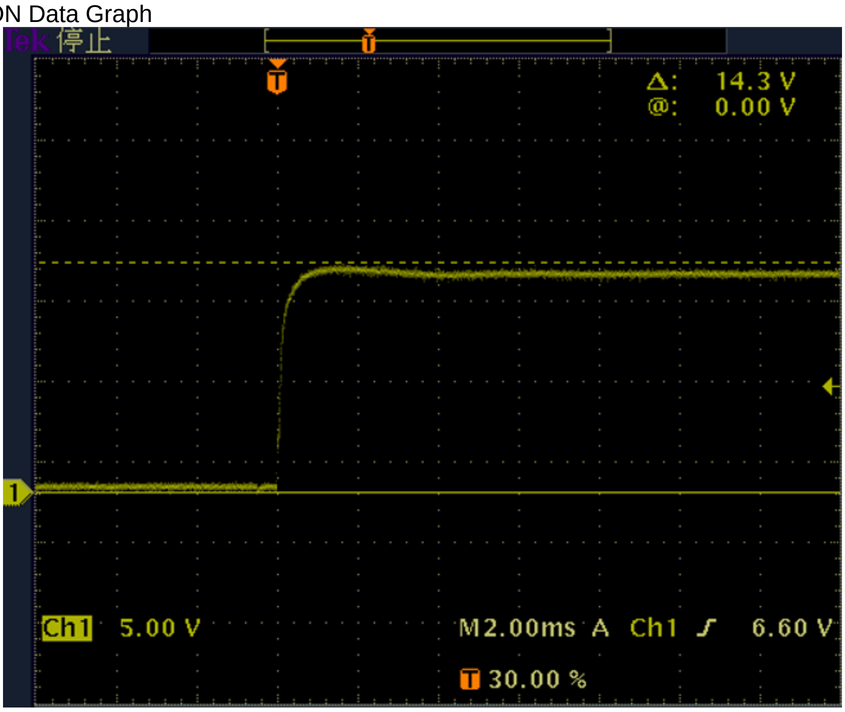
Slow Pulse: OFF-->ON Data Graph



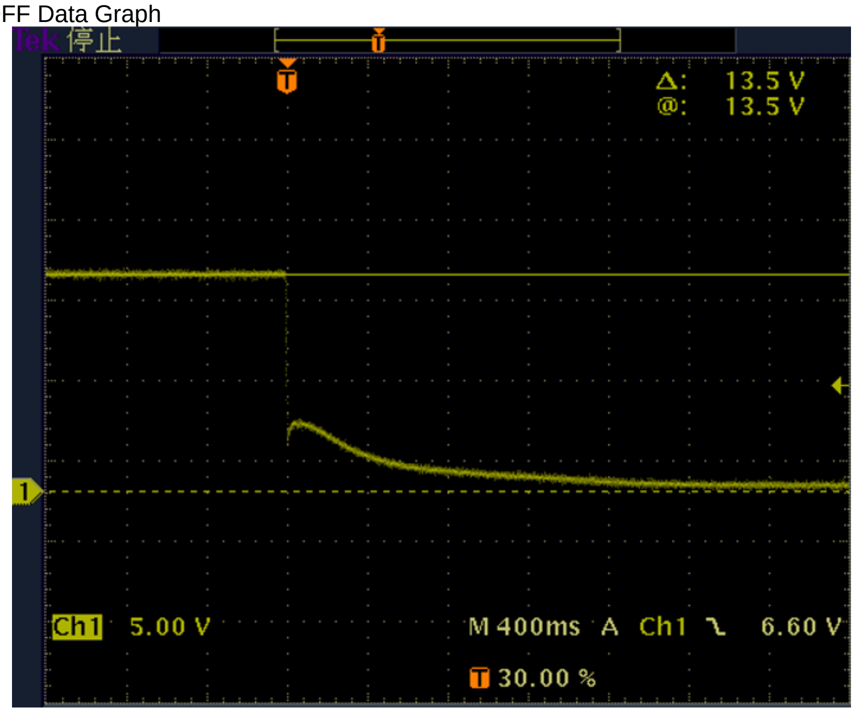
Slow Pulse: ON-->OFF Data Graph



Fast Pulse: OFF-->ON Data Graph

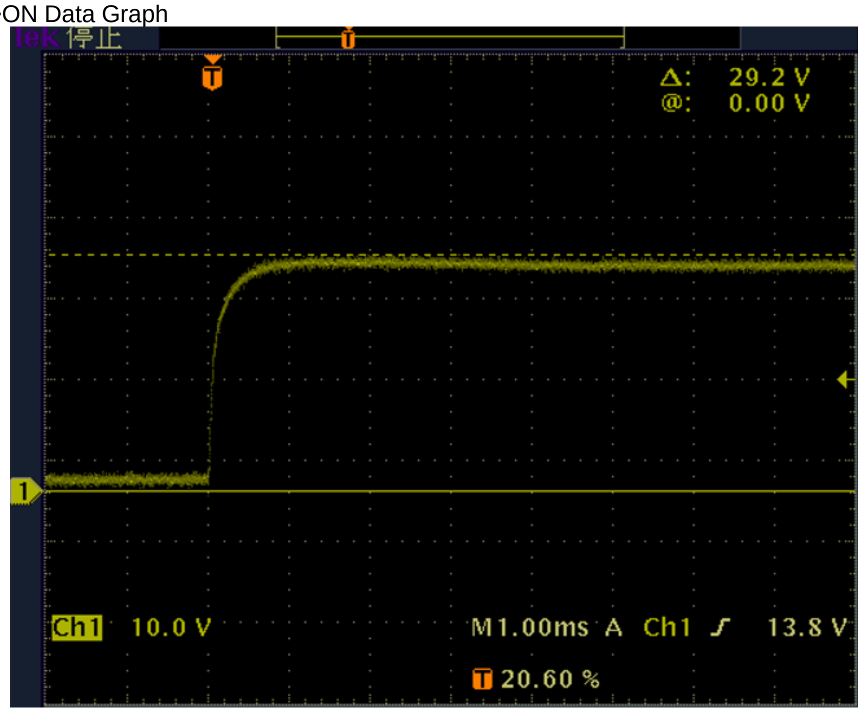


Fast Pulse: ON-->OFF Data Graph

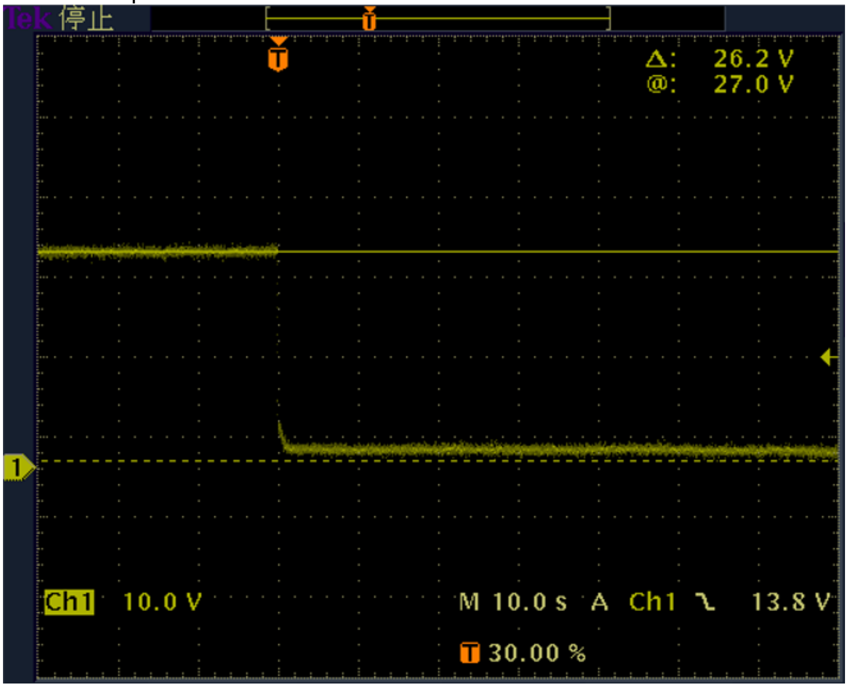


Vehicles with 24 V systems

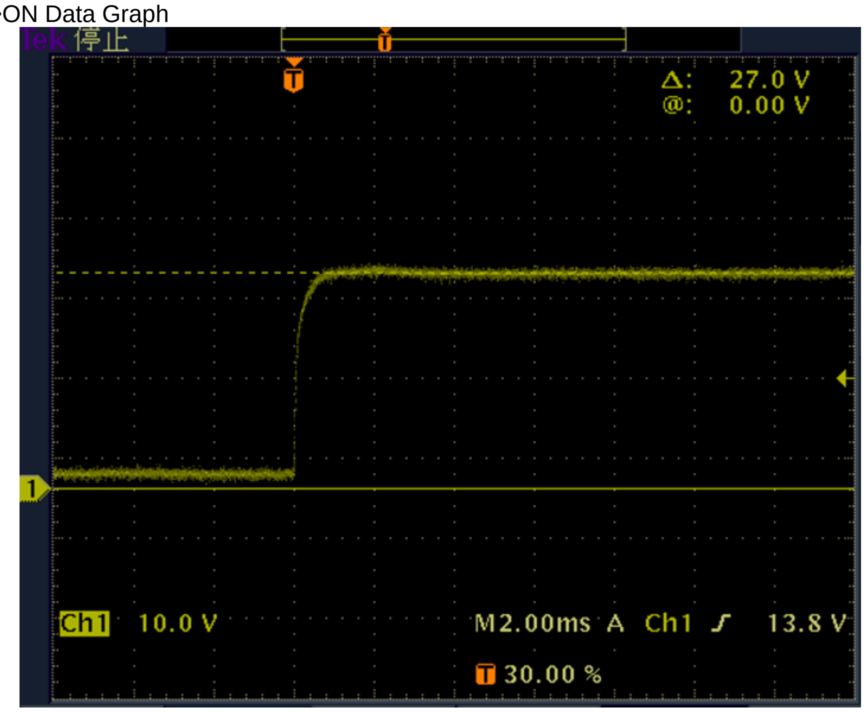
Slow Pulse: OFF-->ON Data Graph



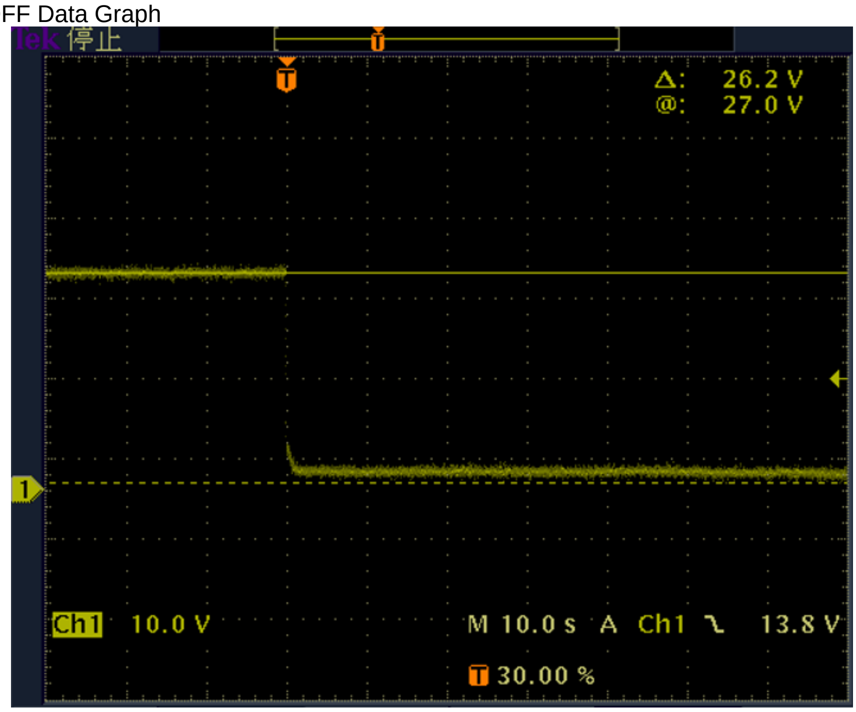
Slow Pulse: ON-->OFF Data Graph



Fast Pulse: OFF-->ON Data Graph



Fast Pulse: ON-->OFF Data Graph



Information document no. VG814-00 relating to type-approval of an electronic subassembly with respect to electromagnetic compatibility (ECE Regulation 10.06 to Supplement 1)

Type:	**VG814**
Total number of pages:	**24**
Date:	**24/02/2022**

INDEX

GENERAL

1. Make (trade name of manufacturer):
inhand

2. Type: **VG814**

General commercial description(s):
InVehicle Gateway

3. Means of identification of type, if marked on the component/separate technical unit [(a)]:
Approval mark

3.1 Location of that marking:
Stuck on the shells, See Drawings of the ESA.

4. Name and address of manufacturer:
Beijing InHand Networks Technology Co., Ltd.
Room 501, 5th floor , building 3, No.18 Yard, Ziyue Road, Chaoyang District, Beijing, China 100102

Name and address of authorised representative, if any:**N/A**

5. In the case of components and separate technical units, location and method of affixing of the ECE approval mark:
Stuck on the shells, See Drawings of the ESA.

6. Address(es) of assembly plant(s):
Inhand Jiaxing Communication Technology Co., Ltd.
318 Ruifeng Street, Gaozhao Street, Xiuzhou District, Jiaxing City, Zhejiang Province, China

7. This ESA shall be approved as a component.

8. Any restrictions of use and conditions for fitting:
N/A

9. Electrical system rated voltage:
DC 12/24V, Negative ground.

10. Charger:
N/A

11. Charging current:
N/A

12. Maximal nominal current (in each mode if necessary) :
N/A

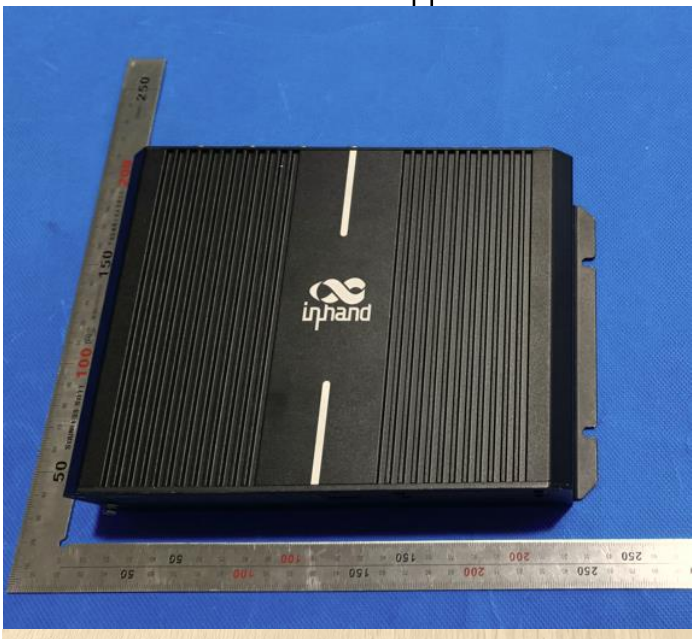
13. Nominal charging voltage:
N/A

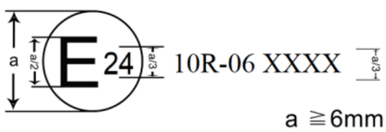
14. Basic ESA interface functions:
N/A

15. Minimum Rsce value (see paragraph 7.11. of this Regulation):
N/A

Drawings of the ESA

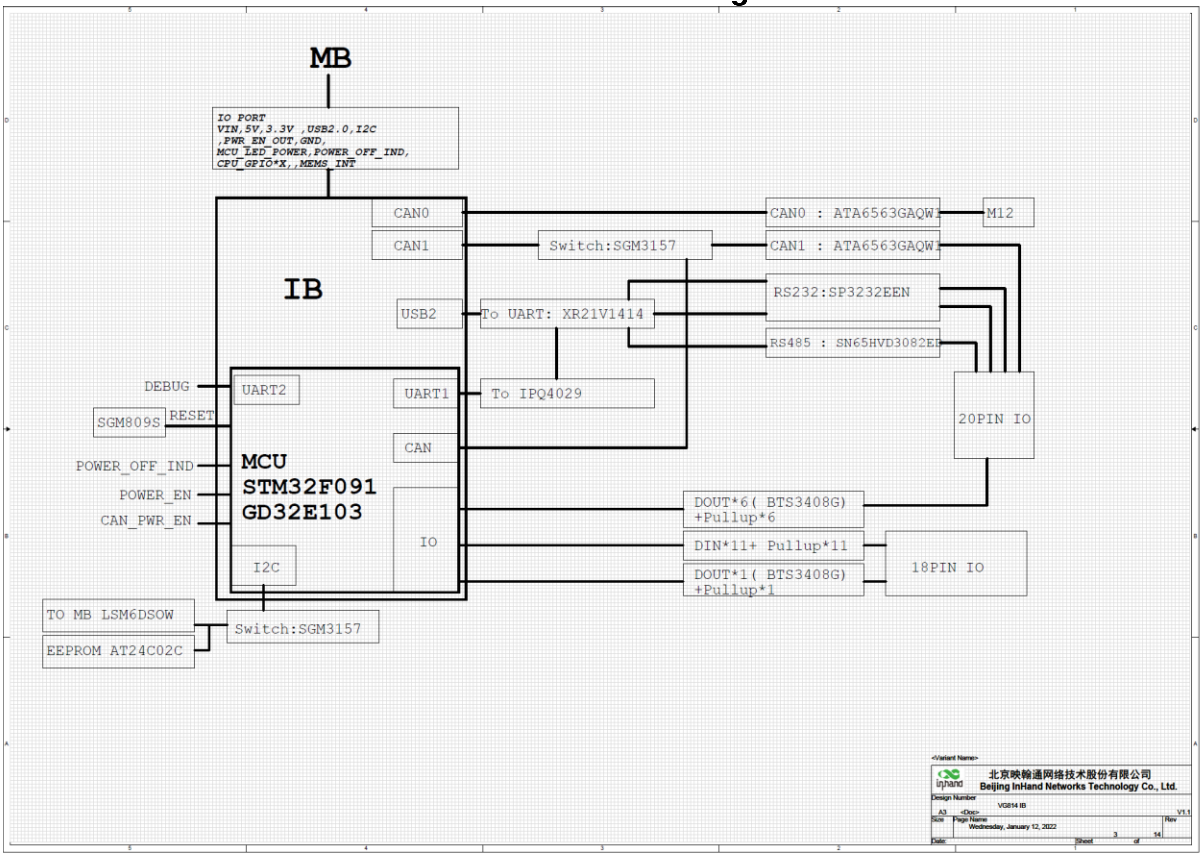
Location of the ECE approval mark

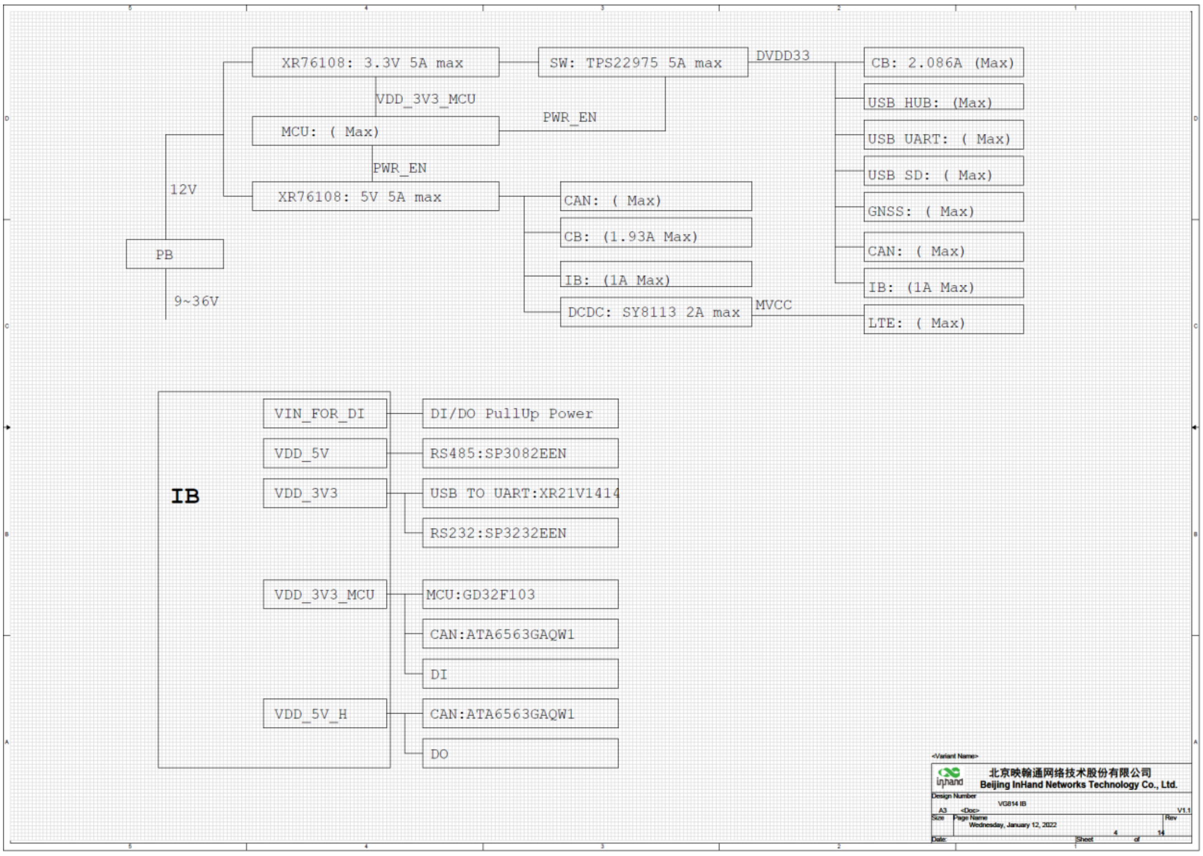


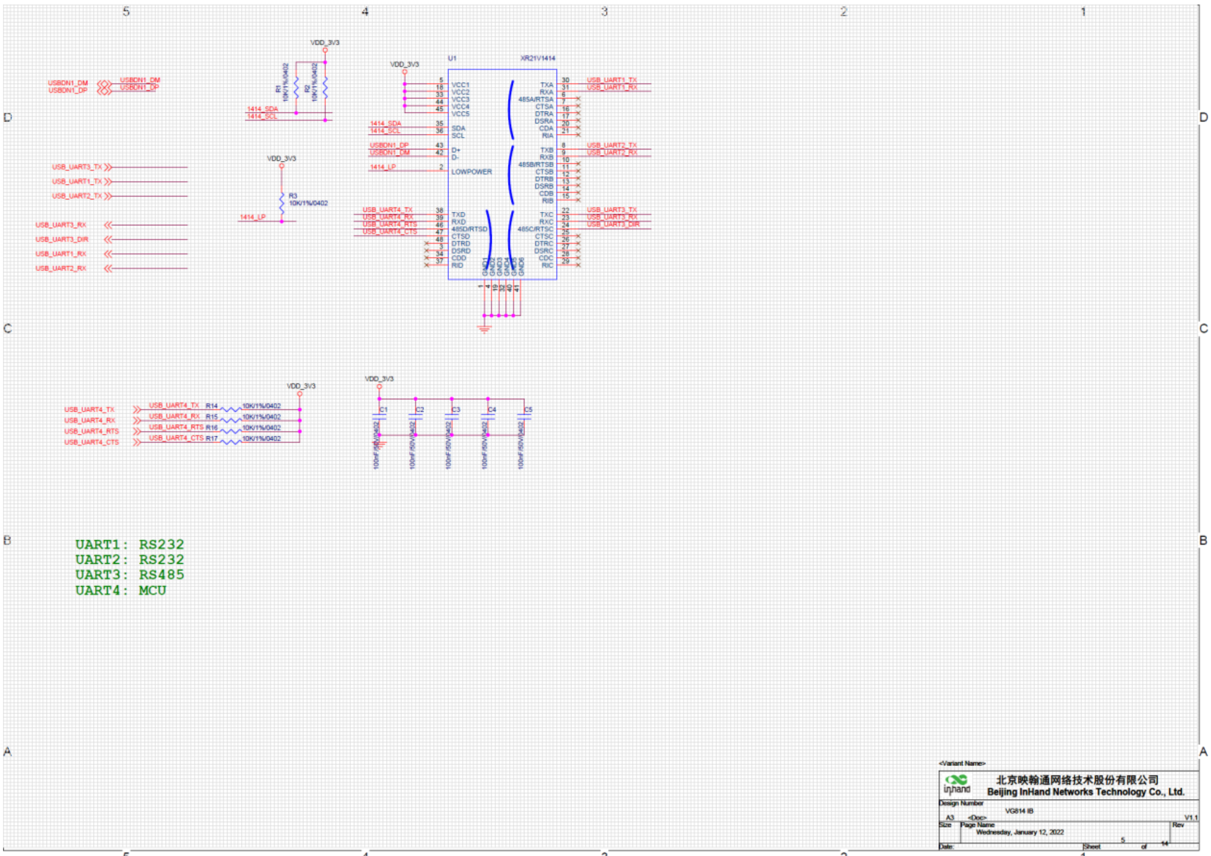


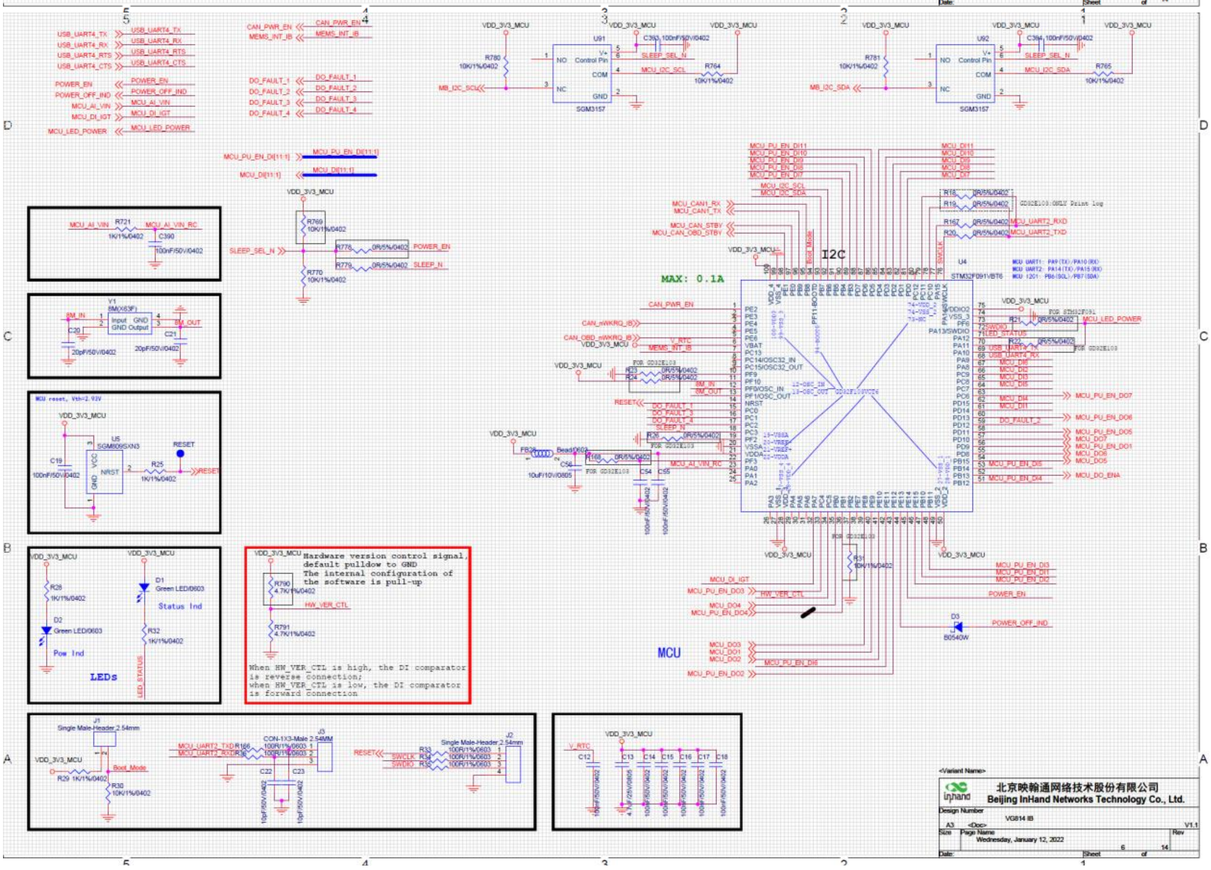
a ≧6mm

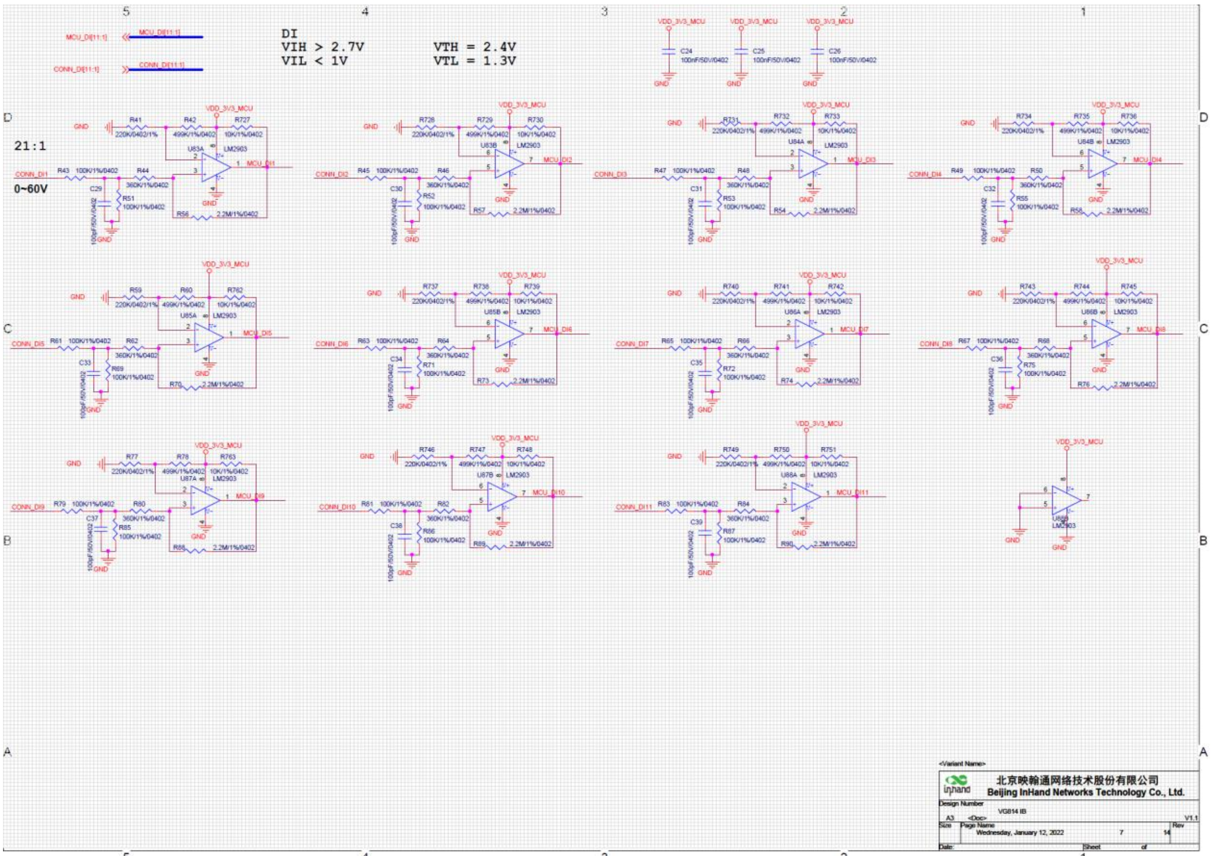
Electronic Circuit Diagram

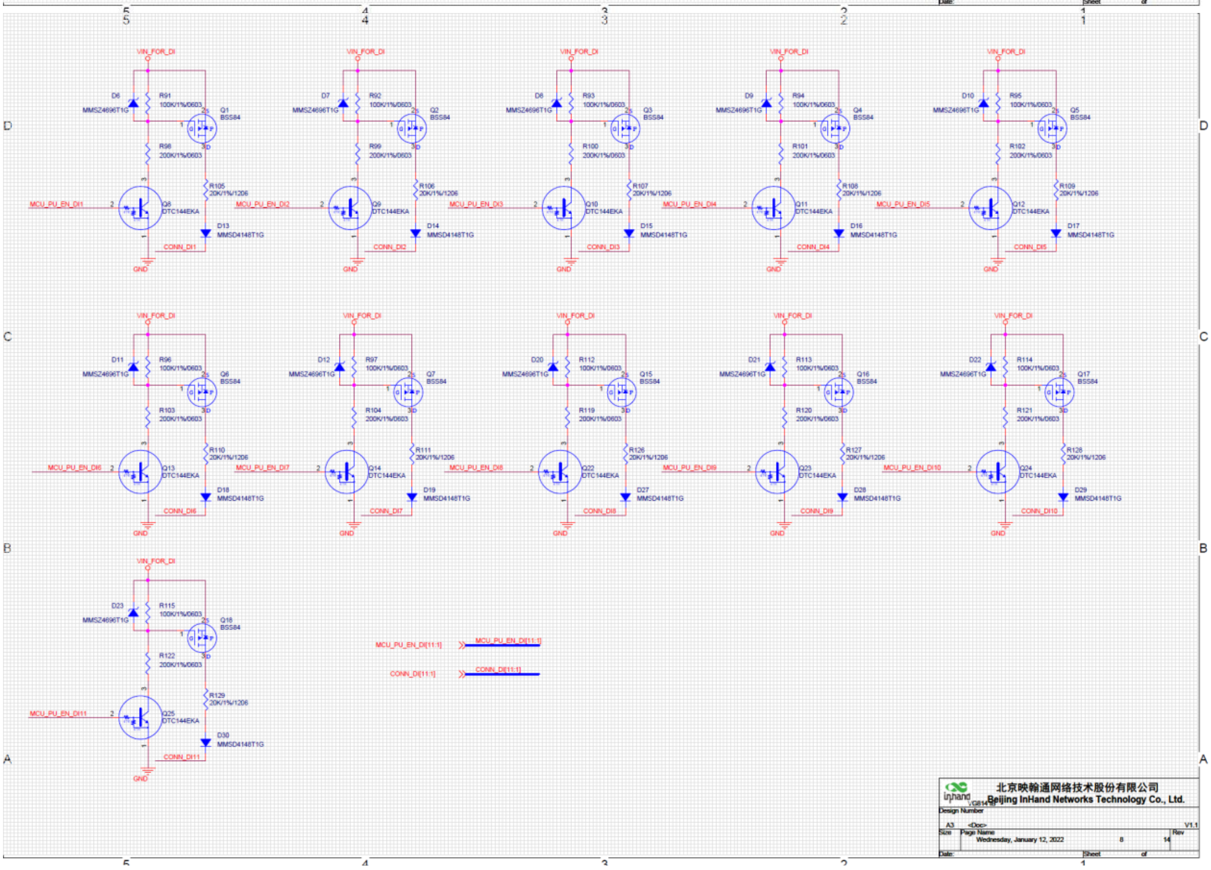


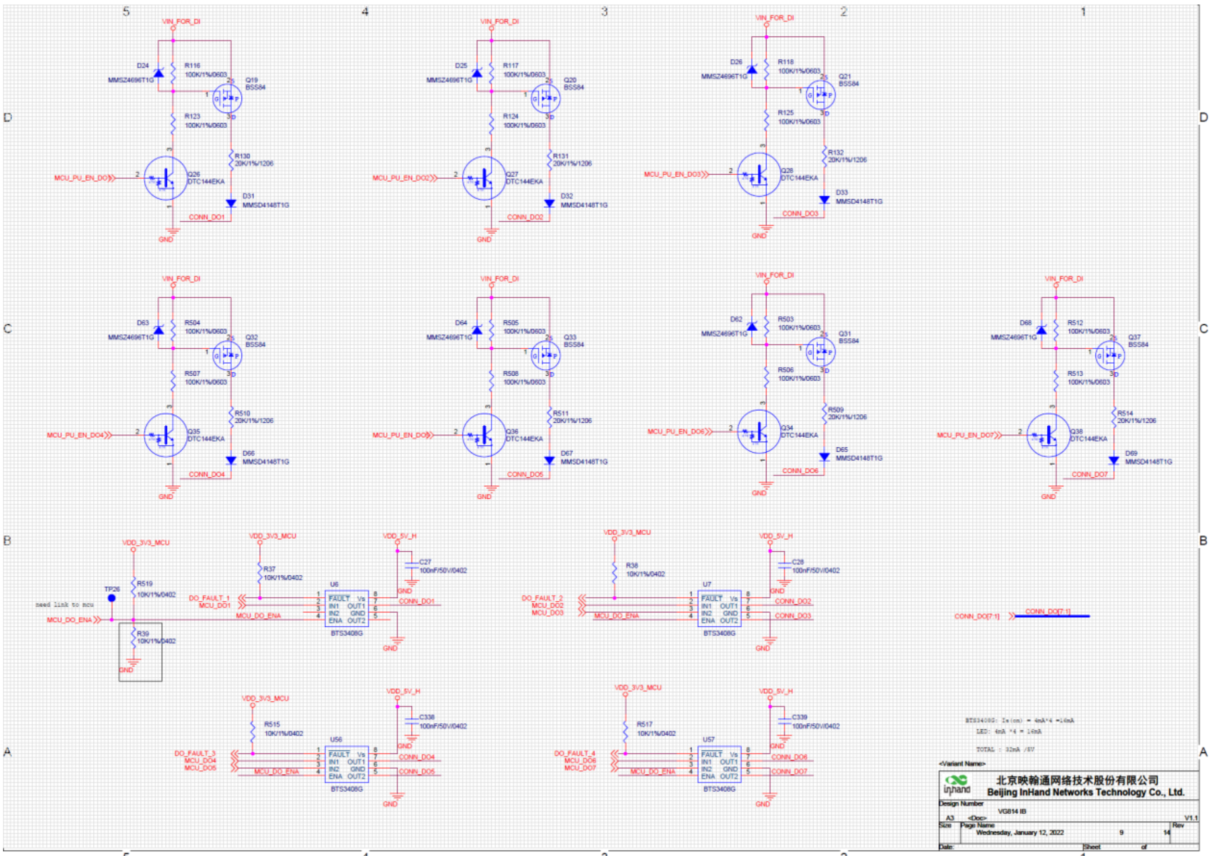


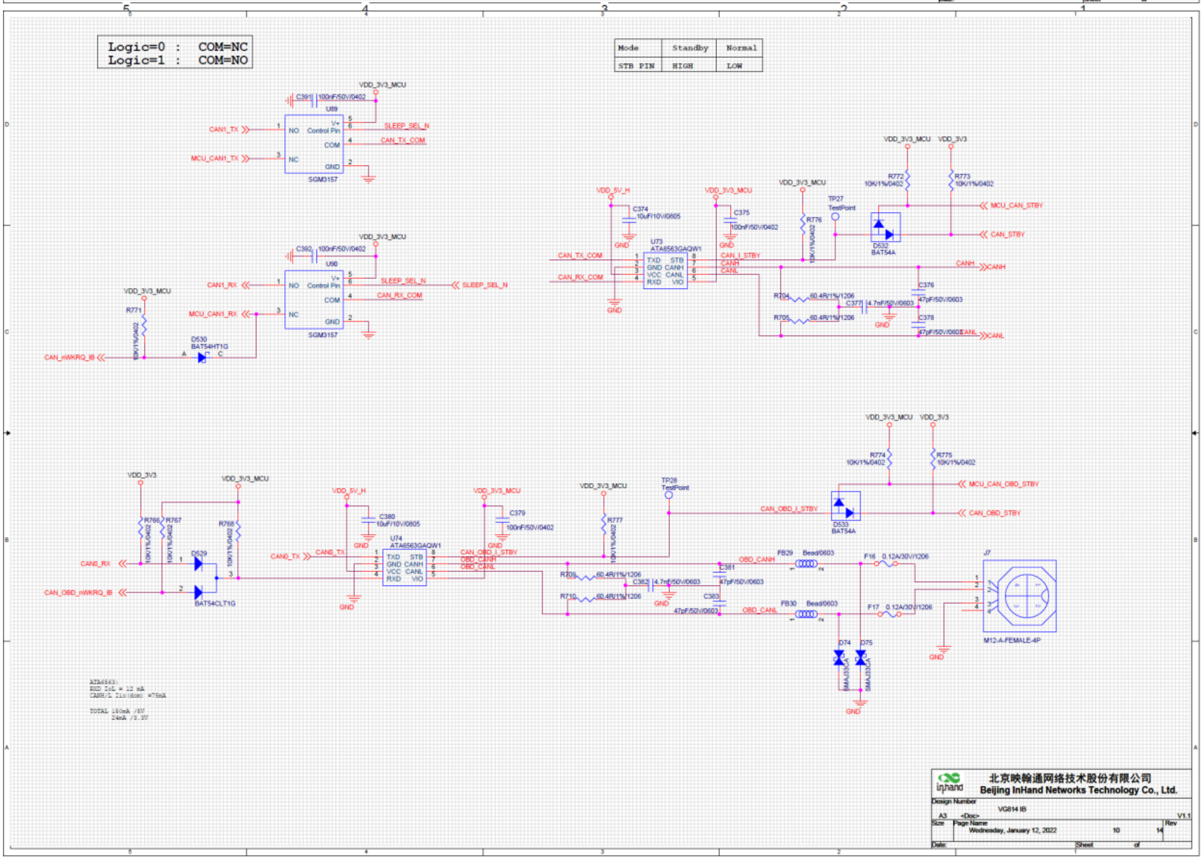

UART1: RS232
UART2: RS232
UART3: RS485
UART4: MCU
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.

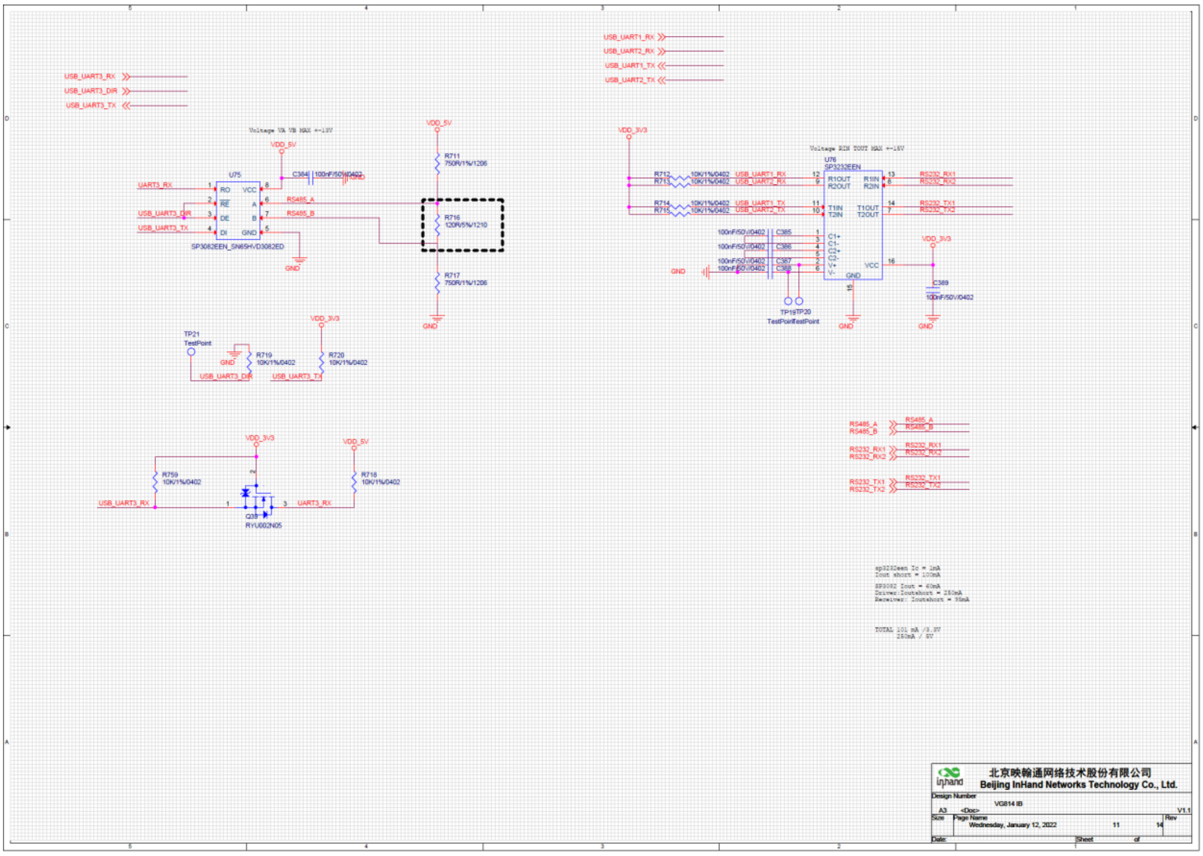

MCU
LEDs
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.

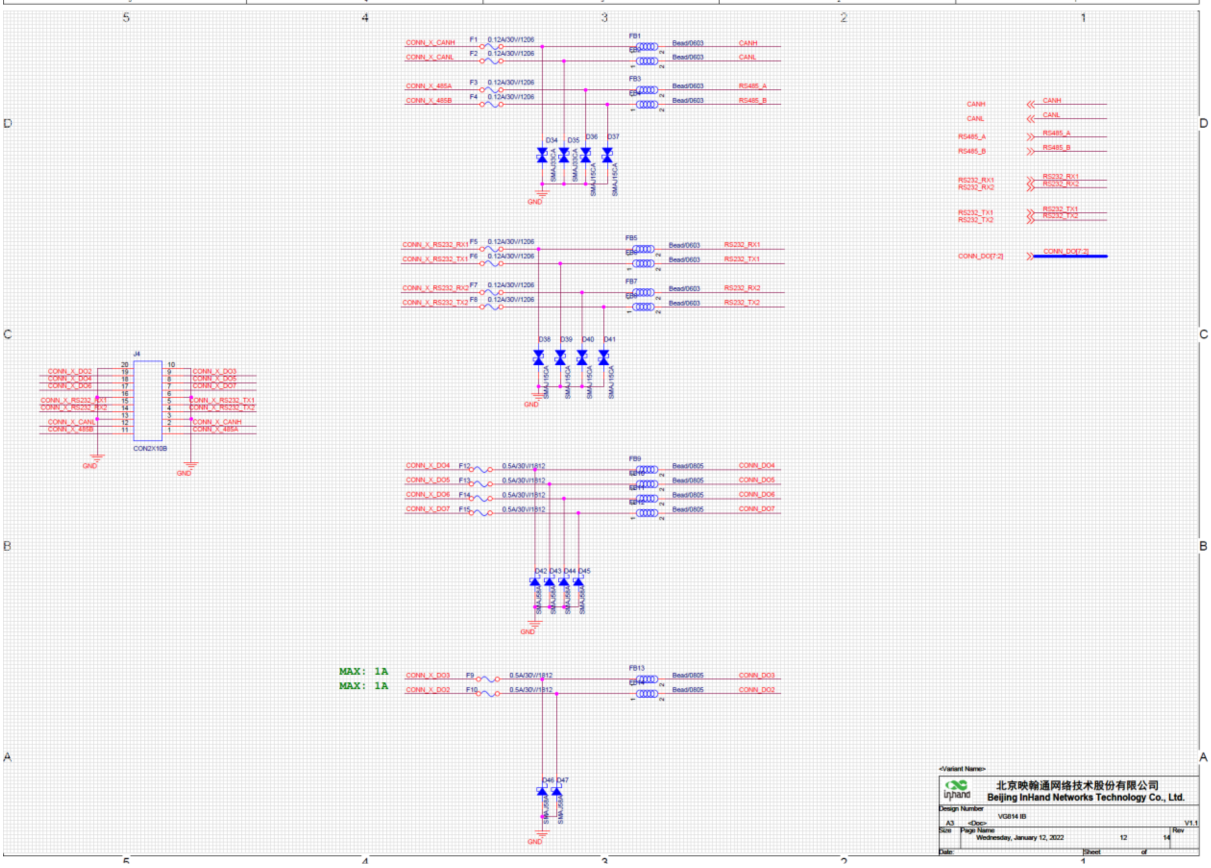


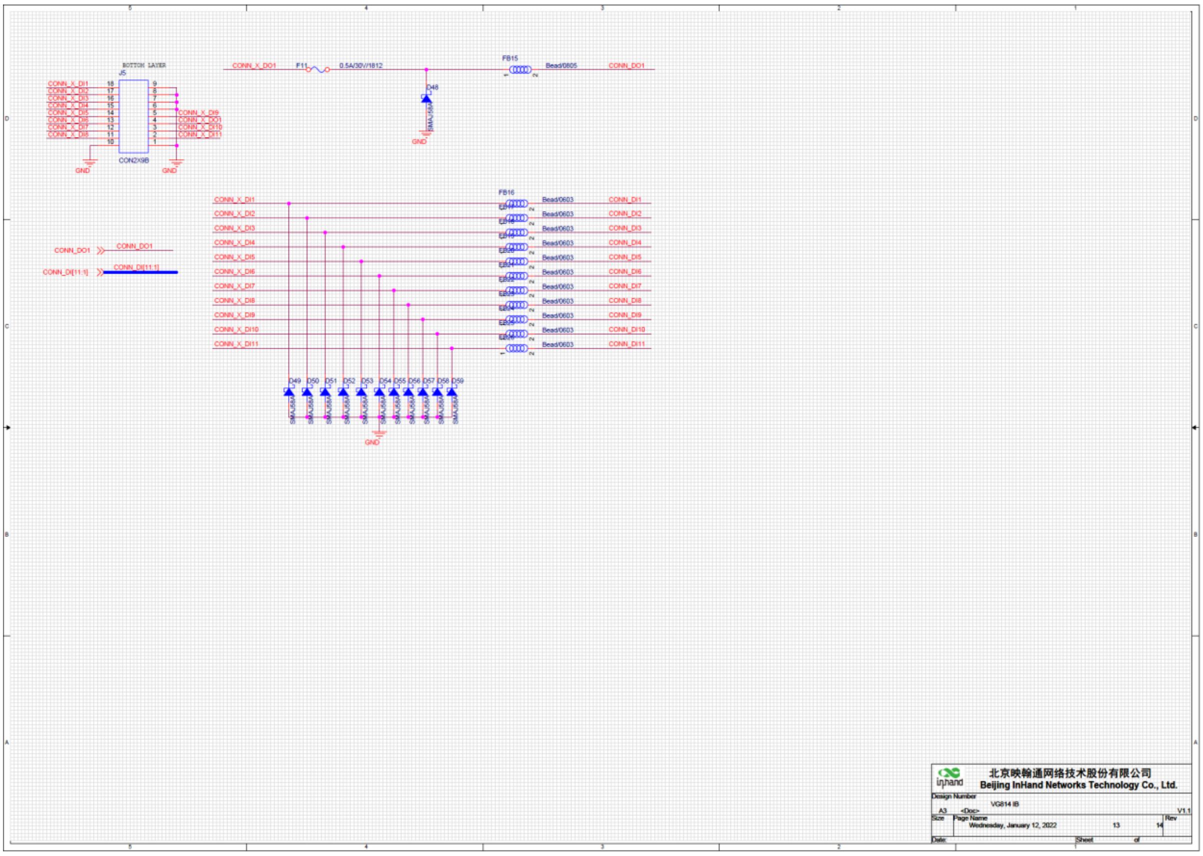








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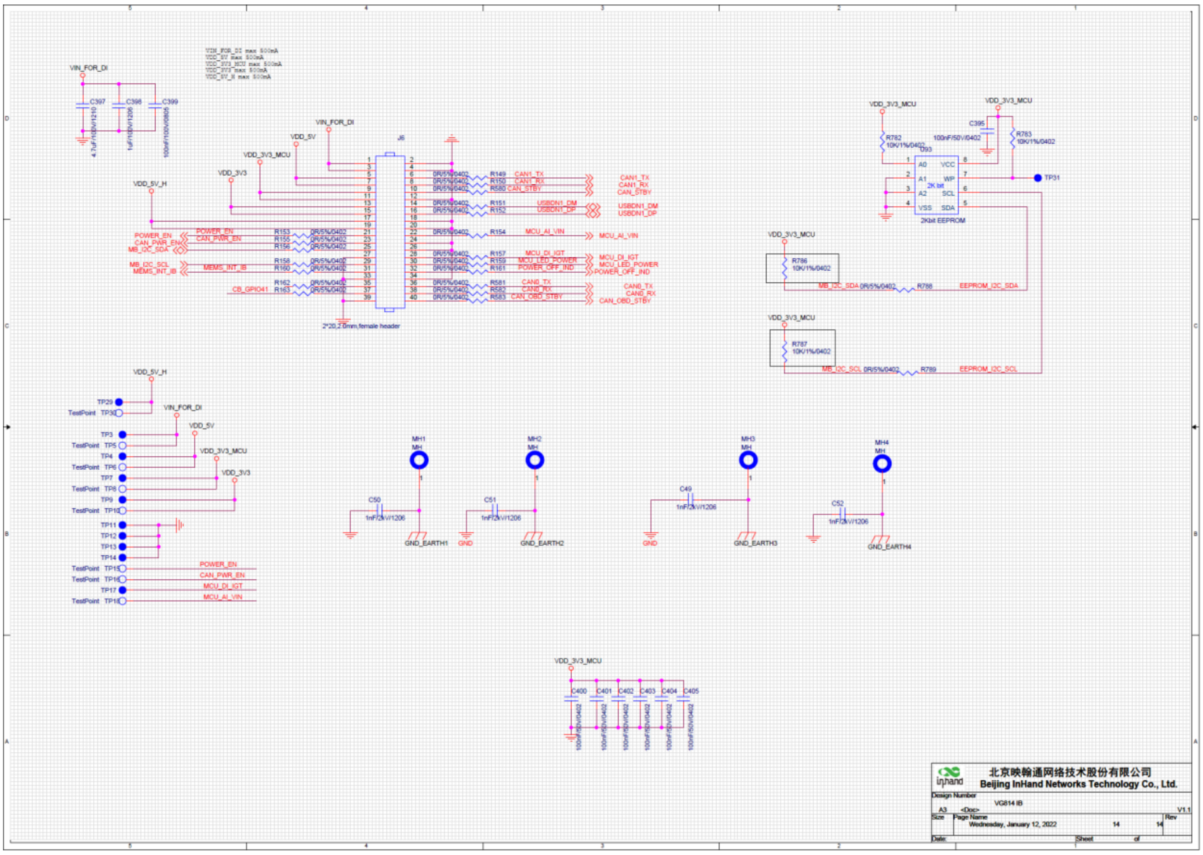
Beijing InHand Networks Technology Co., Ltd.
VG814 IB
Wednesday, January 12, 2022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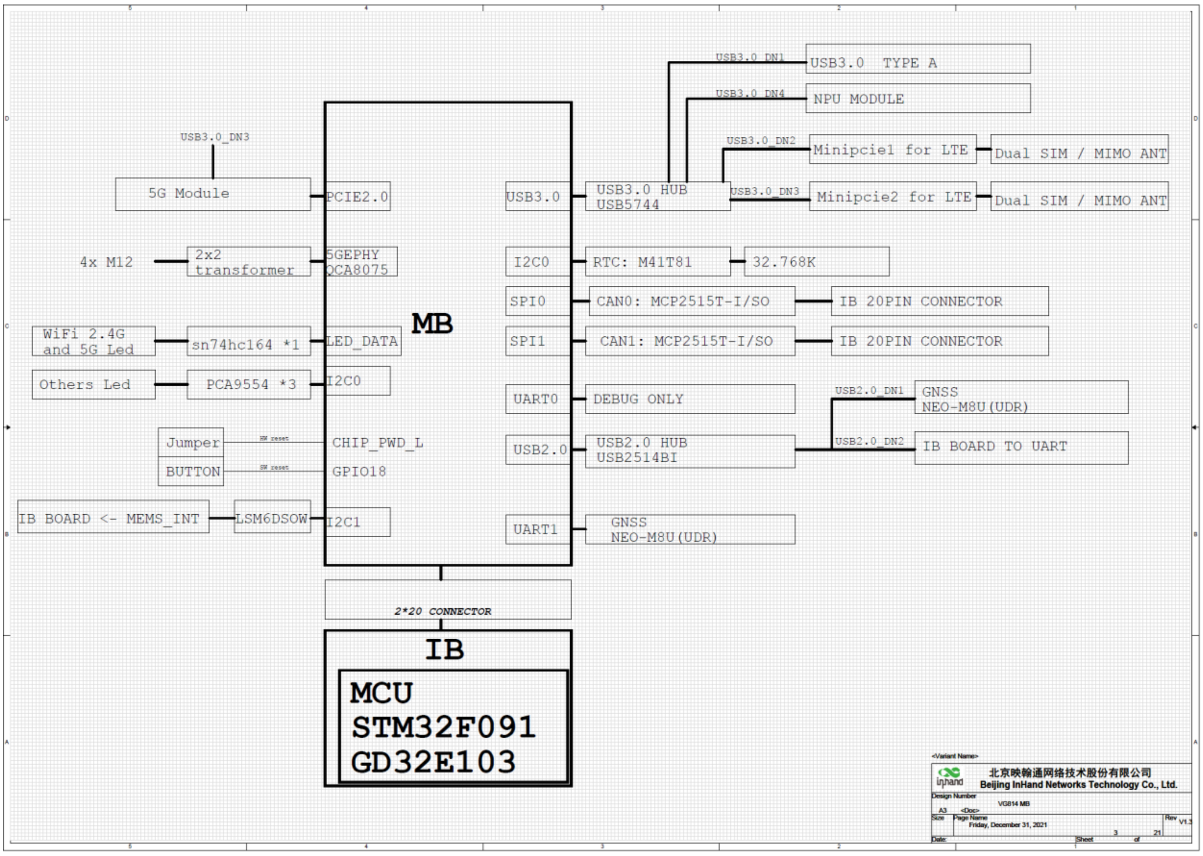
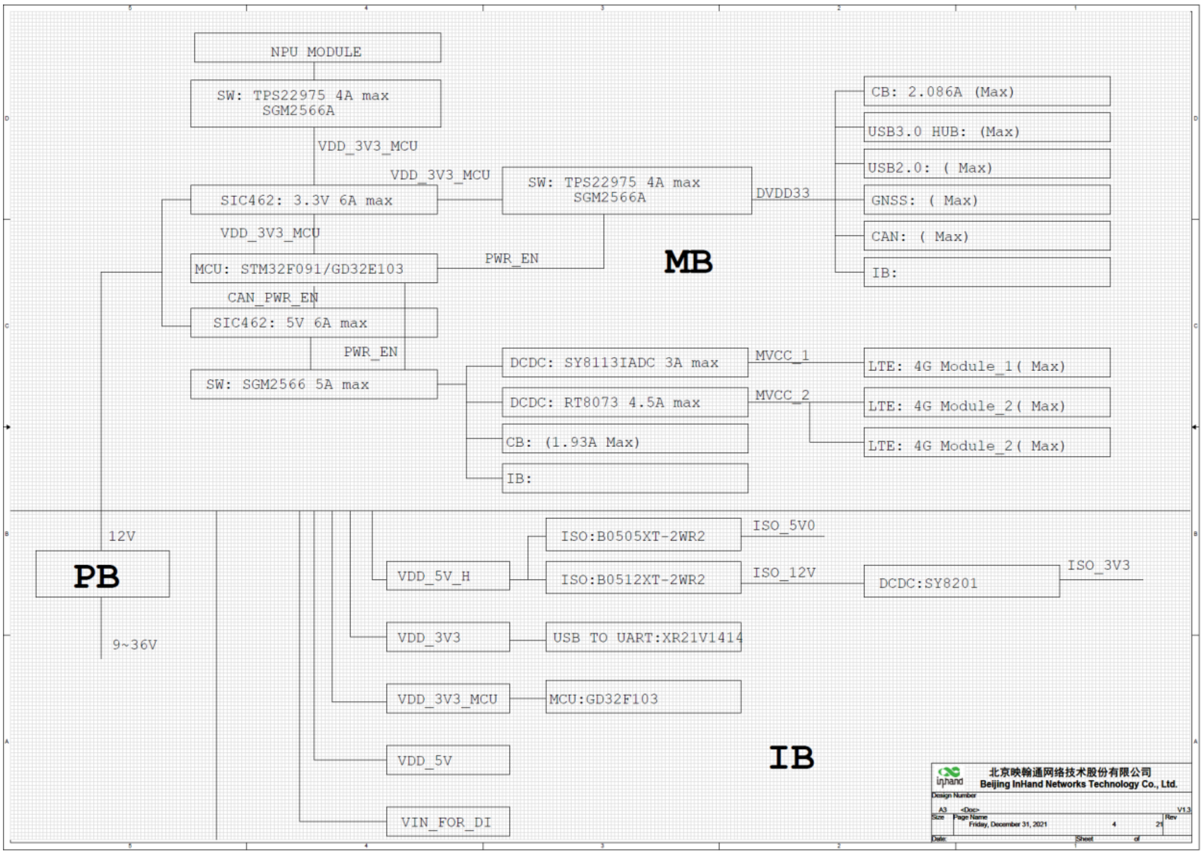
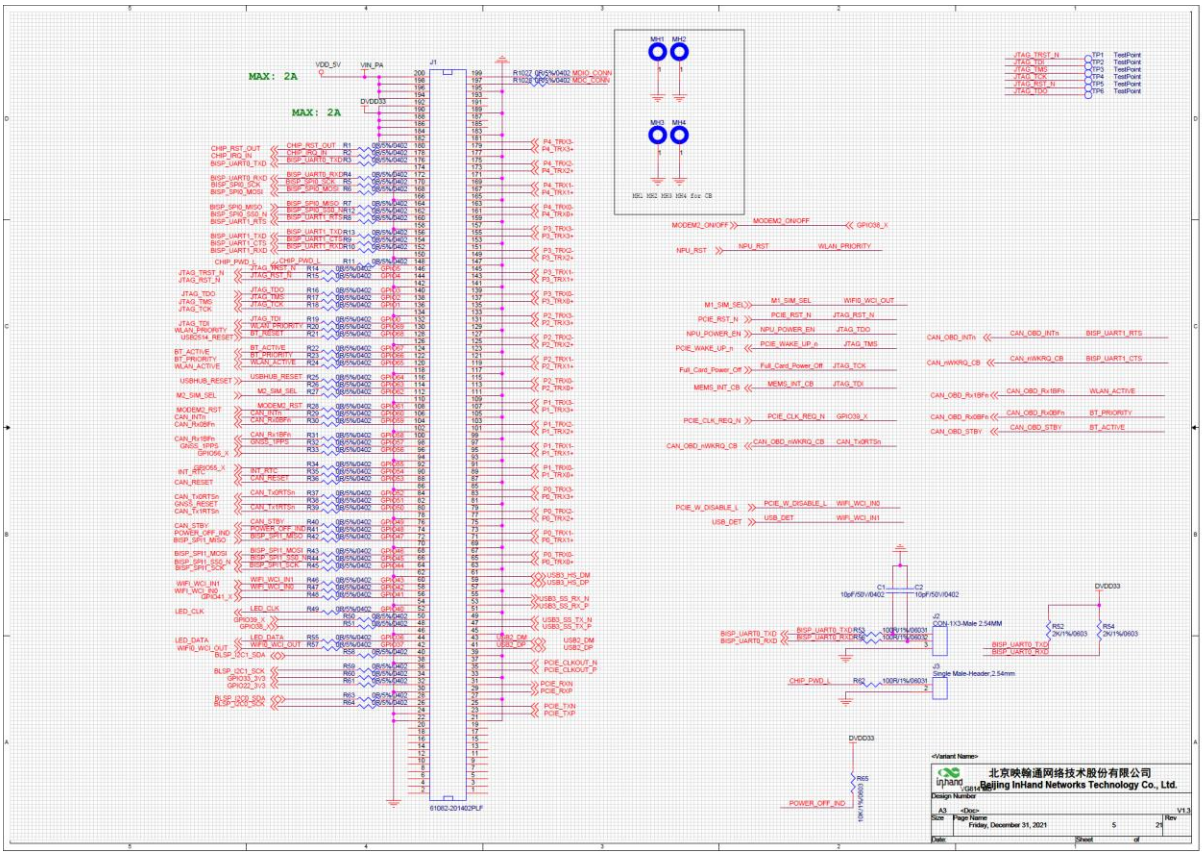
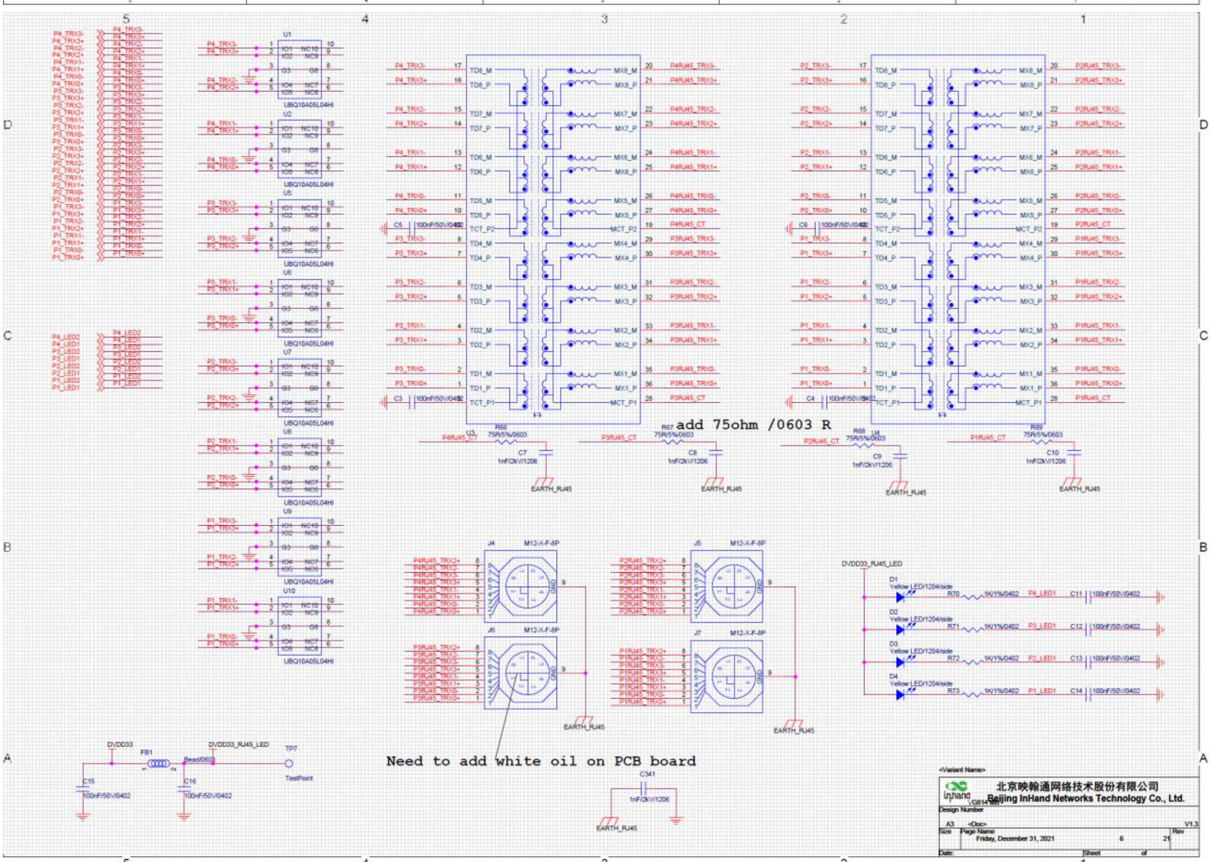
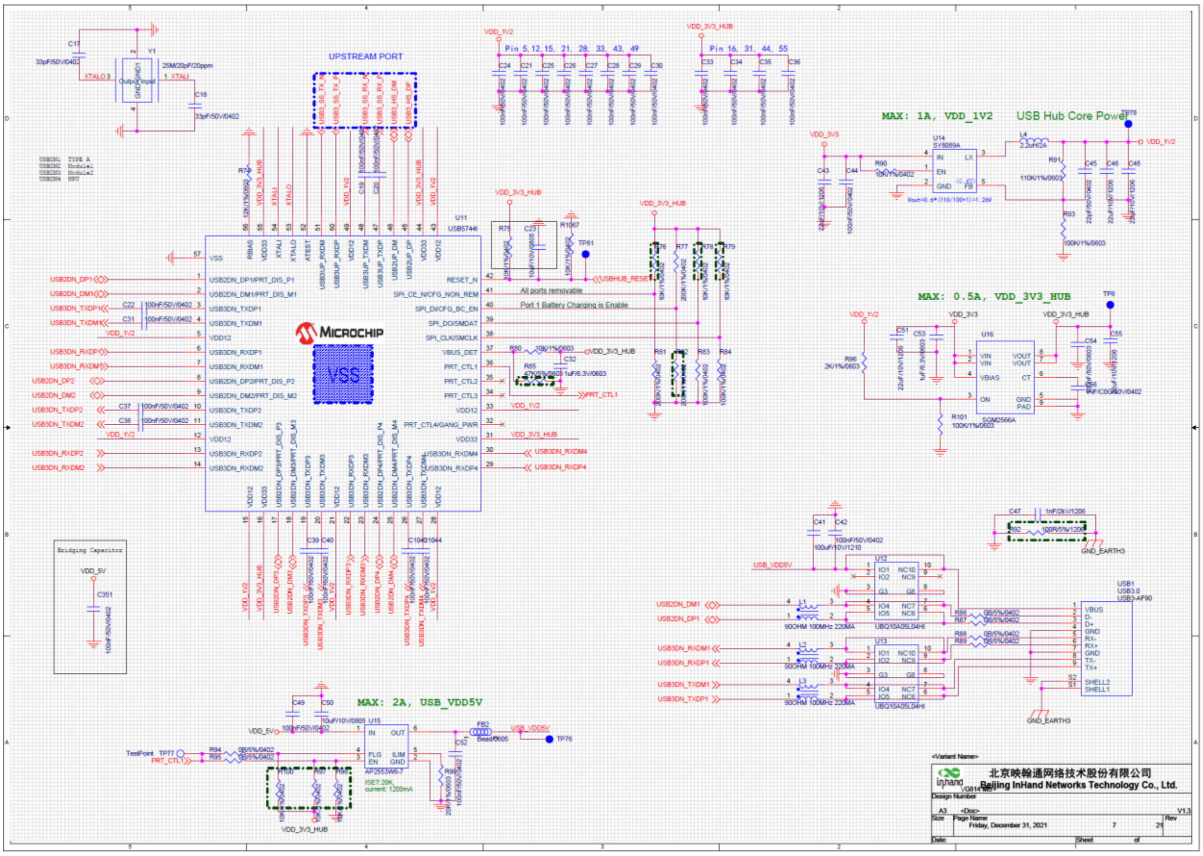
Beijing InHand Networks Technology Co., Ltd.
VG814 IB
Wednesday, January 12, 2022

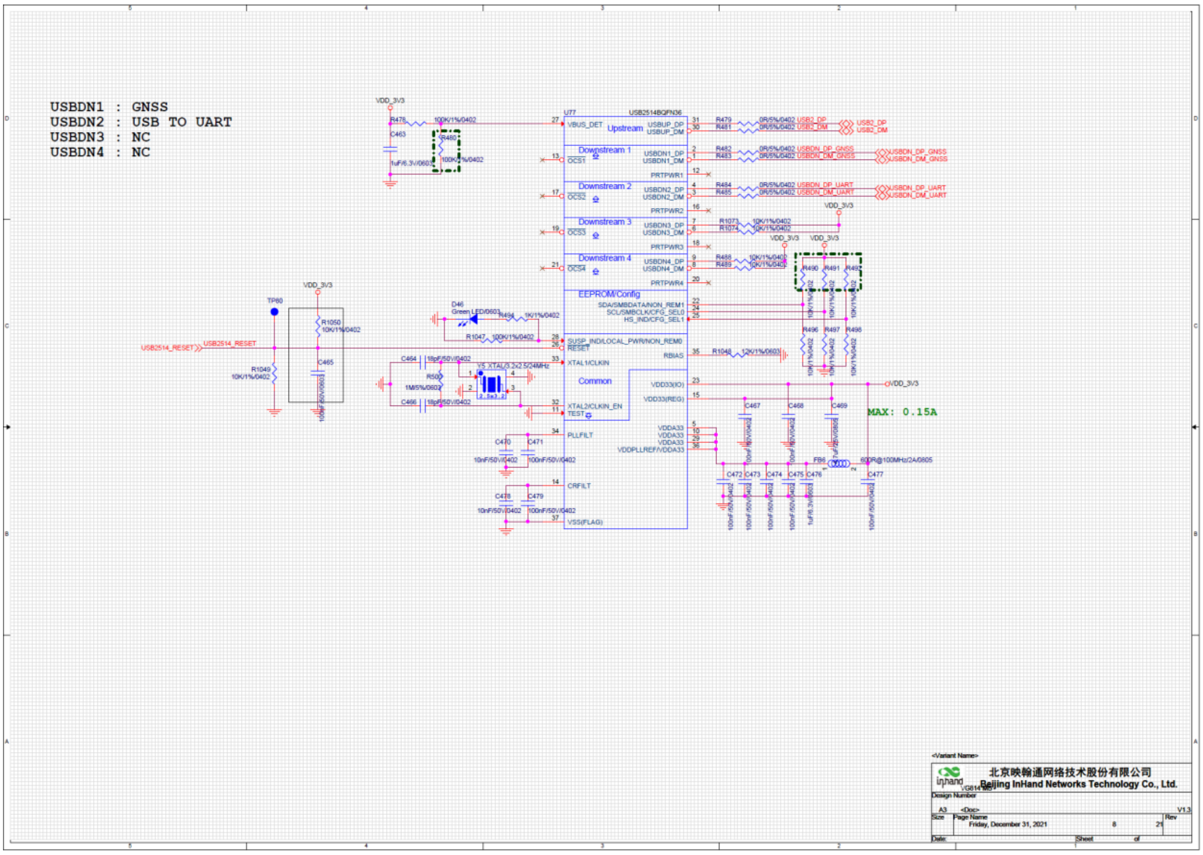

USB3.0 TYPE A
NPU MODULE
Minipcie1 for LTE
Dual SIM / MIMO ANT
Minipcie2 for LTE
Dual SIM / MIMO ANT
5G Module
PCIE2.0
USB3.0
USB3.0 HUB
USB5744
4x M12
2x2 transformer
5GEPHY
QCA8075
I2C0
RTC: M41T81
32.768K
SPI0
CAN0: MCP2515T-I/SO
IB 20PIN CONNECTOR
WiFi 2.4G and 5G Led
sn74hc164 *1
LED_DATA
MB
SPI1
CAN1: MCP2515T-I/SO
IB 20PIN CONNECTOR
Others Led
PCA9554 *3
I2C0
UART0
DEBUG ONLY
GNSS
NEO-M8U(UDR)
Jumper
BUTTON
CHIP_PWD_L
GPIO18
USB2.0
USB2.0 HUB
USB2514BI
IB BOARD TO UART
IB BOARD <- MEMS_INT
LSM6DSOW
I2C1
UART1
GNSS
NEO-M8U(UDR)
2*20 CONNECTOR
IB
MCU
STM32F091
GD32E103
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.

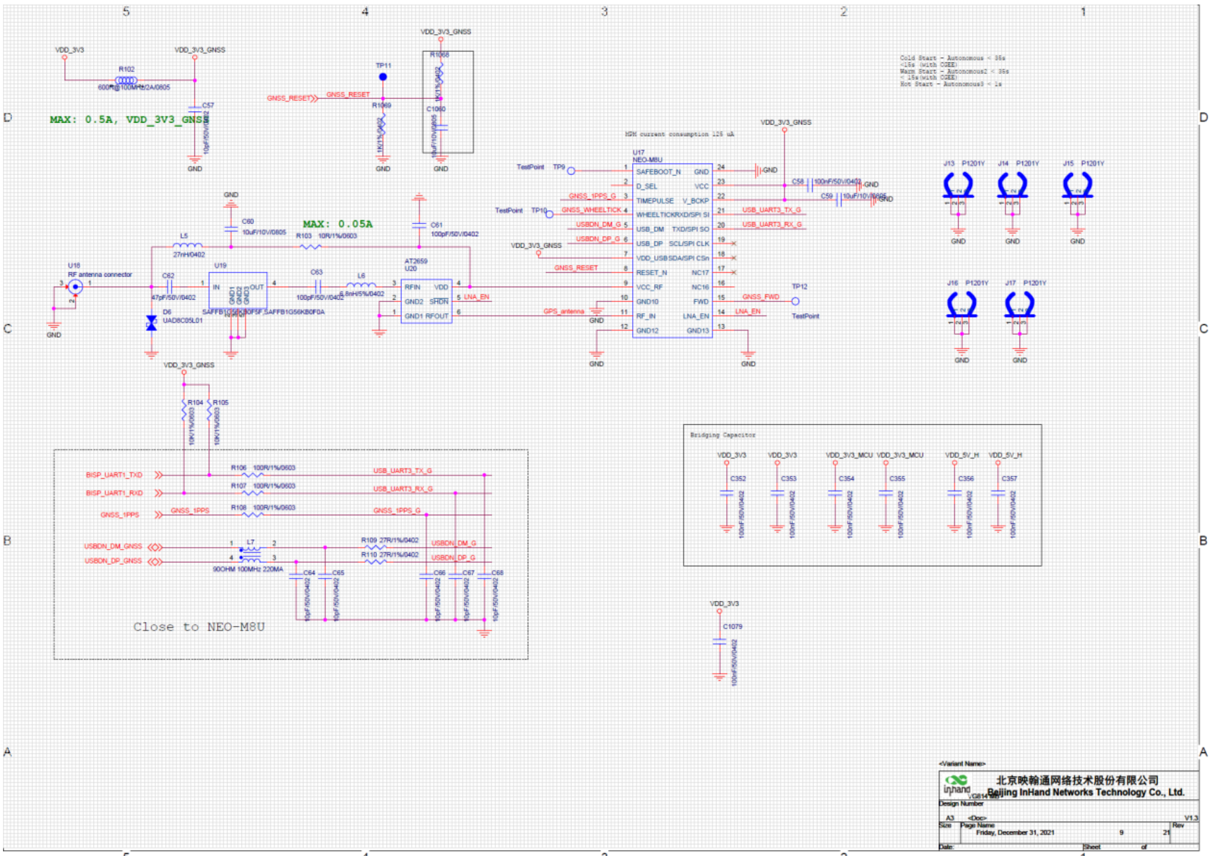

NPU MODULE
SW: TPS22975 4A max
SGM2566A
VDD_3V3_MCU
CB: 2.086A (Max)
USB3.0 HUB: (Max)
USB2.0: (Max)
GNSS: (Max)
CAN: (Max)
IB:
SIC462: 3.3V 6A max
SW: TPS22975 4A max
SGM2566A
DVDD33
MCU: STM32F091/GD32E103
PWR_EN
MB
CAN_PWR_EN
SIC462: 5V 6A max
SW: SGM2566 5A max
DCDC: SY8113IADC 3A max
MVCC_1
LTE: 4G Module_1(Max)
DCDC: RT8073 4.5A max
MVCC_2
LTE: 4G Module_2(Max)
CB: (1.93A Max)
LTE: 4G Module_2(Max)
12V
PB
9~36V
ISO:B0505XT-2WR2
ISO_5V0
VDD_5V_H
ISO:B0512XT-2WR2
ISO_12V
DCDC:SY8201
ISO_3V3
VDD_3V3
USB TO UART:XR21V1414
VDD_3V3_MCU
MCU:GD32F103
VDD_5V
IB
VIN_FOR_DI
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.

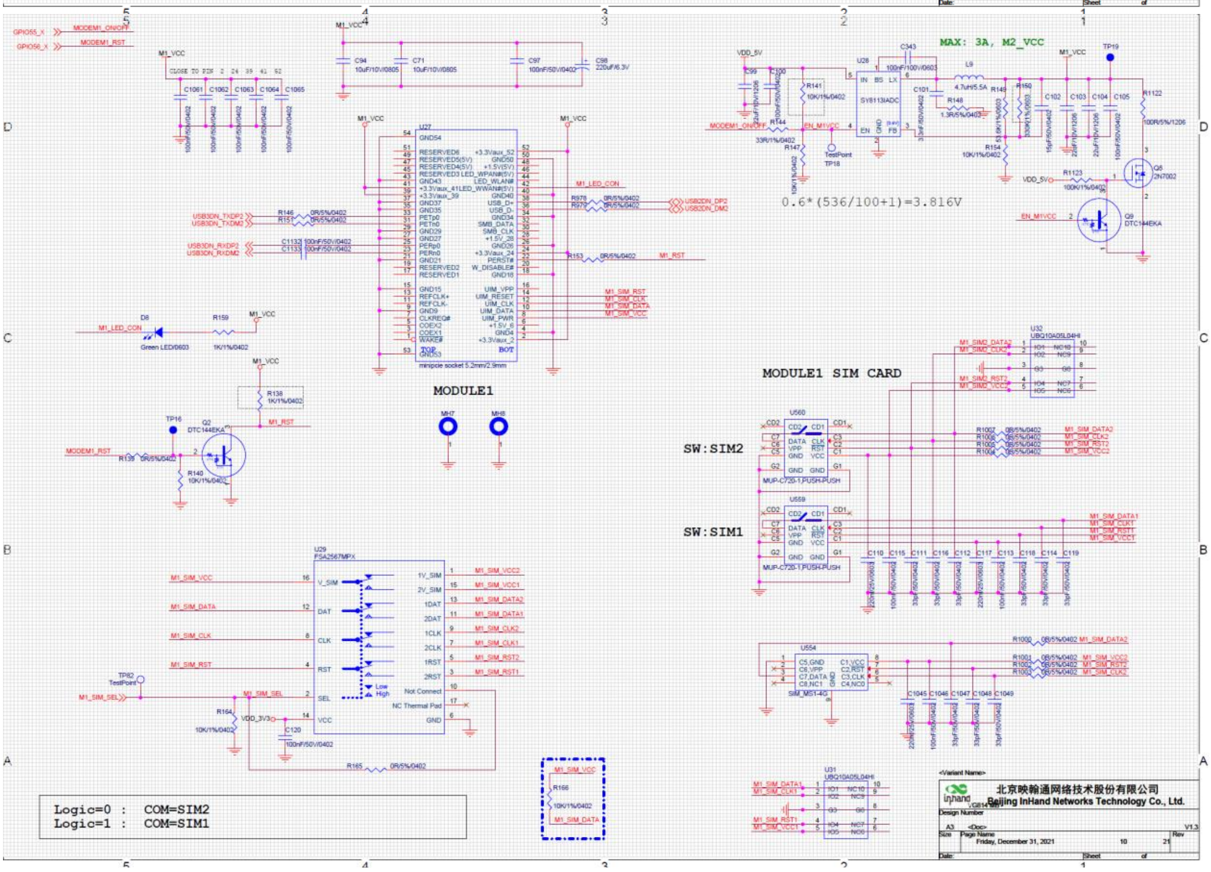

MAX: 2A
MAX: 2A
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.

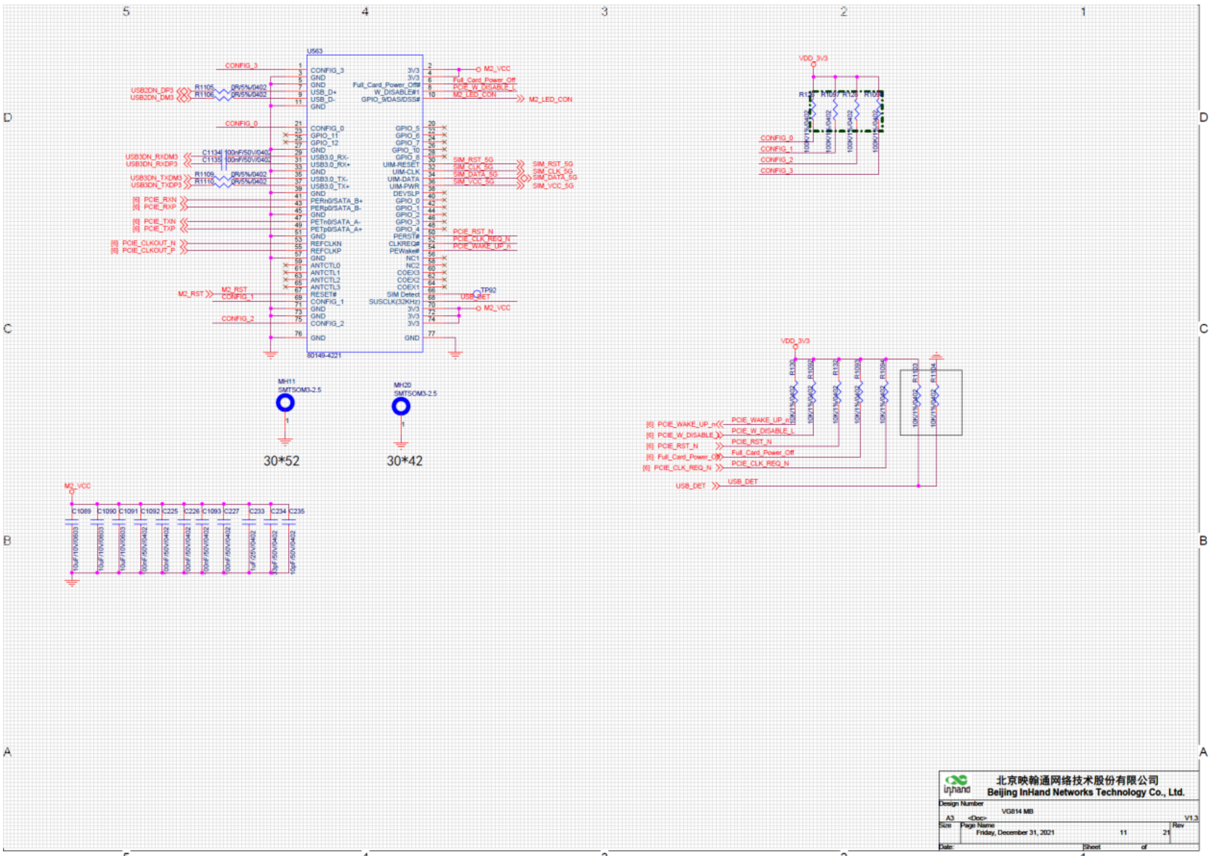

add 75ohm /0603 R
Need to add white oil on PCB board
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.

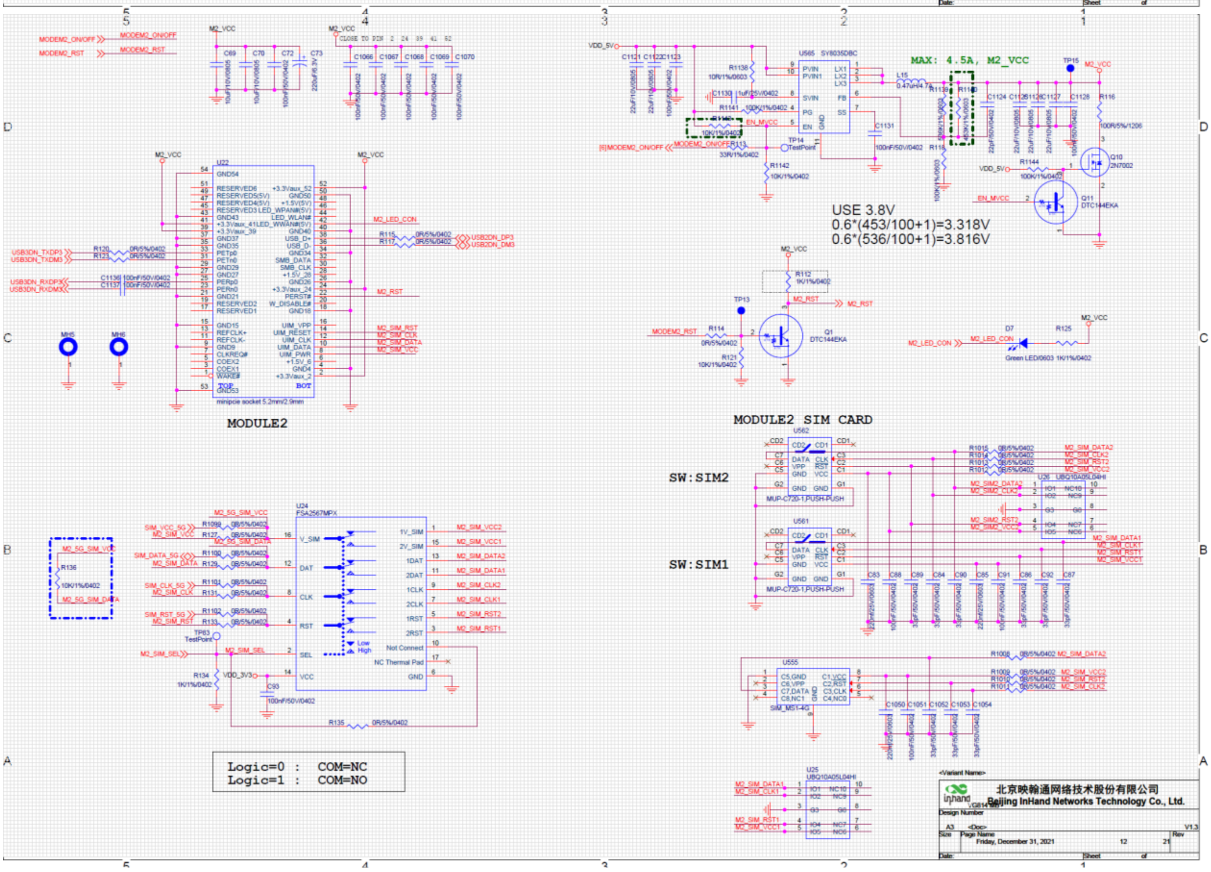

UPSTREAM PORT
MICROCHIP
VSS
MAX: 1A, VDD_1V2
USB Hub Core Power
MAX: 0.5A, VDD_3V3_HUB
MAX: 2A, USB_VDD5V
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.

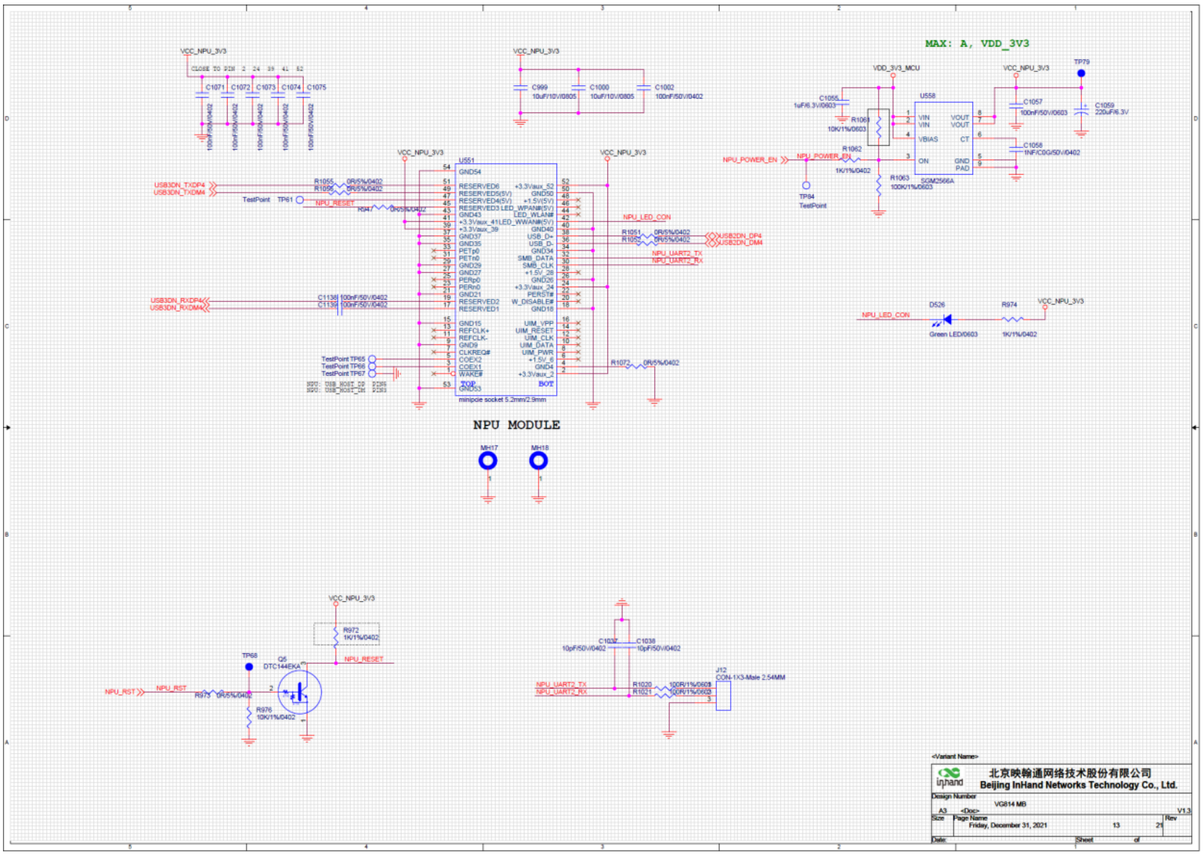

USBDN1 : GNSS
USBDN2 : USB TO UART
USBDN3 : NC
USBDN4 : NC
MAX: 0.15A
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.

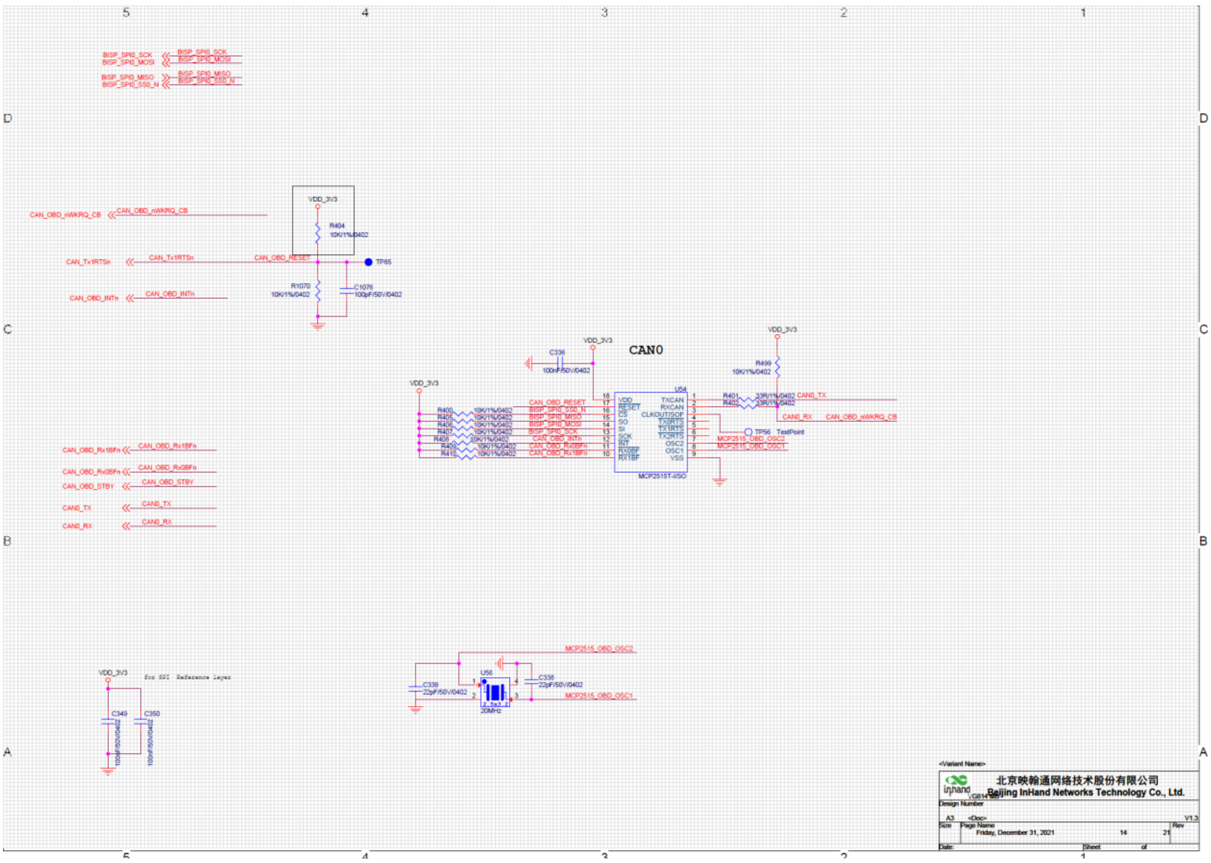

MAX: 0.5A, VDD_3V3_GNSS
MAX: 0.05A
Close to NEO-M8U
Bridging Capacitor
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.
Friday, December 31, 2021

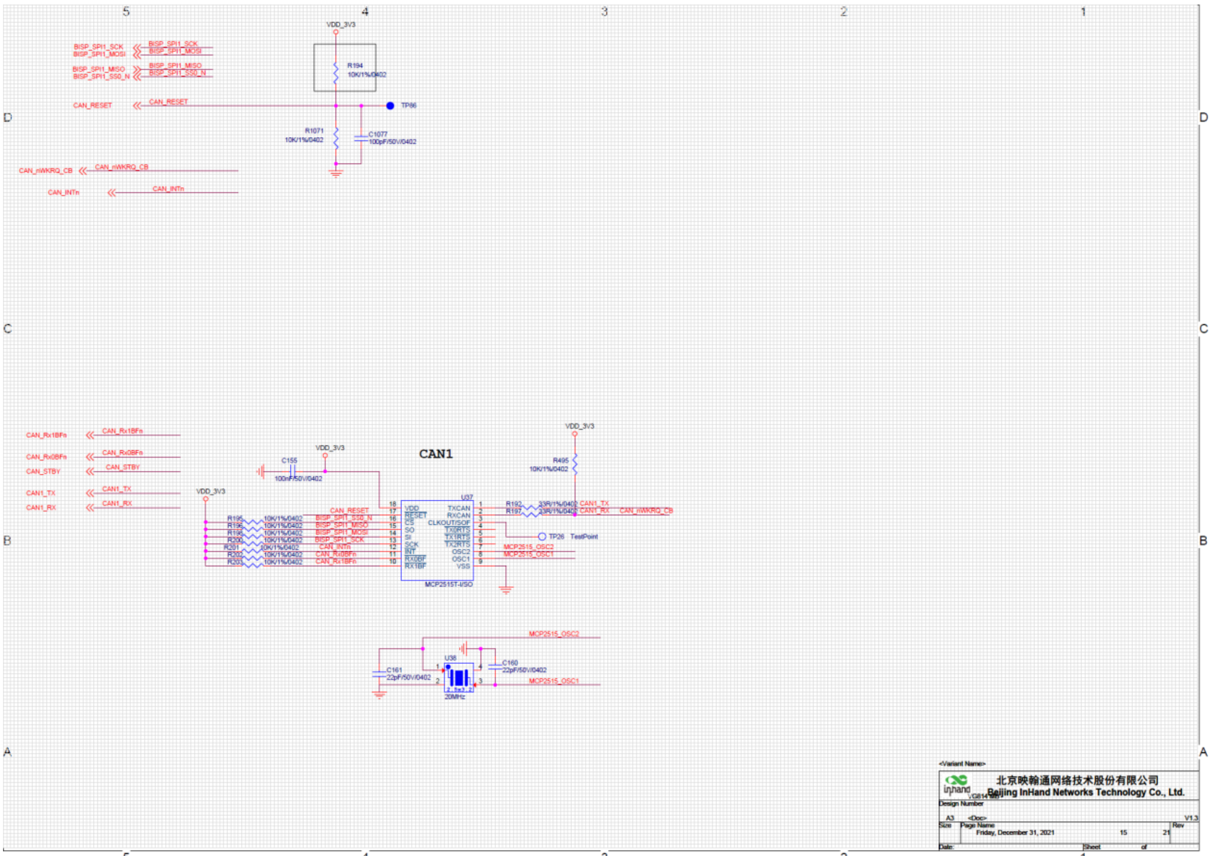

MAX: 3A, M2_VCC
0.6*(536/100+1)=3.816V
MODULE1
MODULE1 SIM CARD
SW:SIM2
SW:SIM1
Logic=0 : COM=SIM2
Logic=1 : COM=SIM1
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.
Friday, December 31, 2021

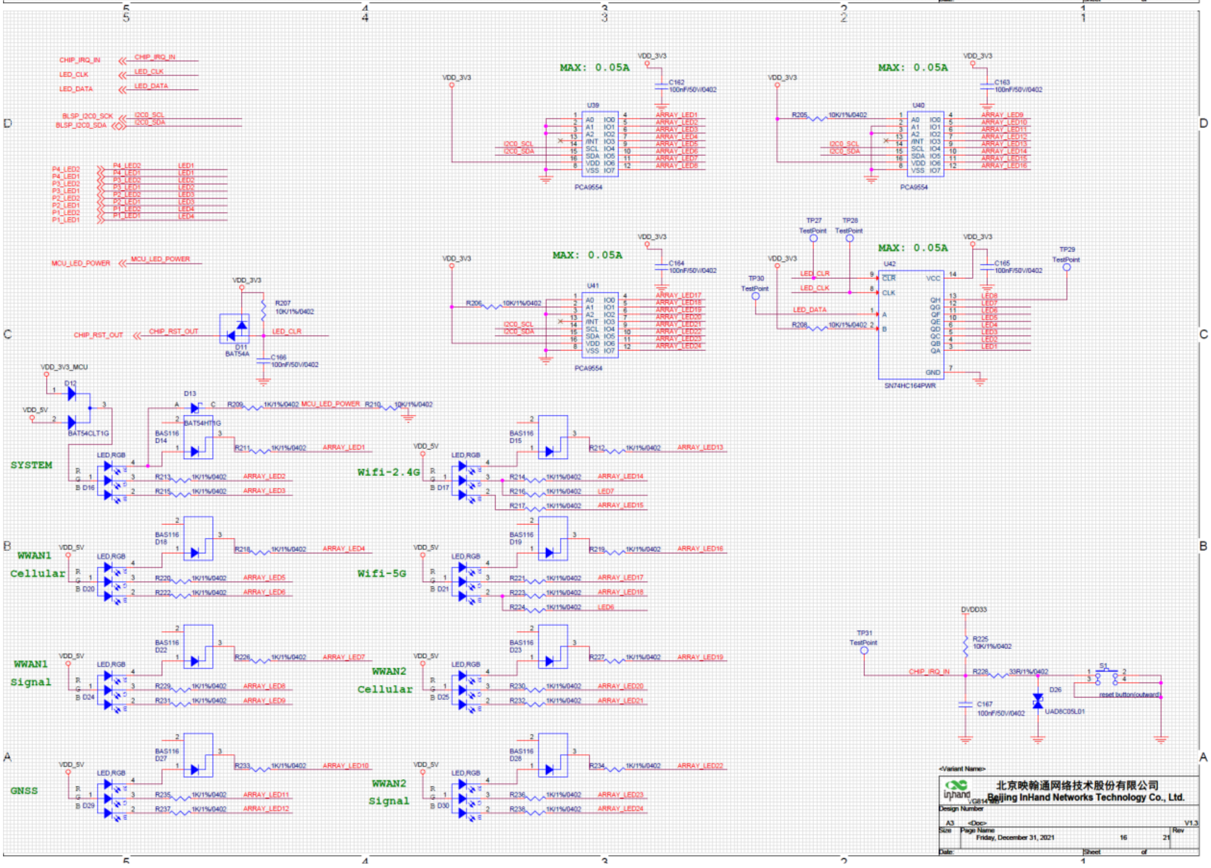

30*52
30*42
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.
VG814 MB
Friday, December 31, 2021

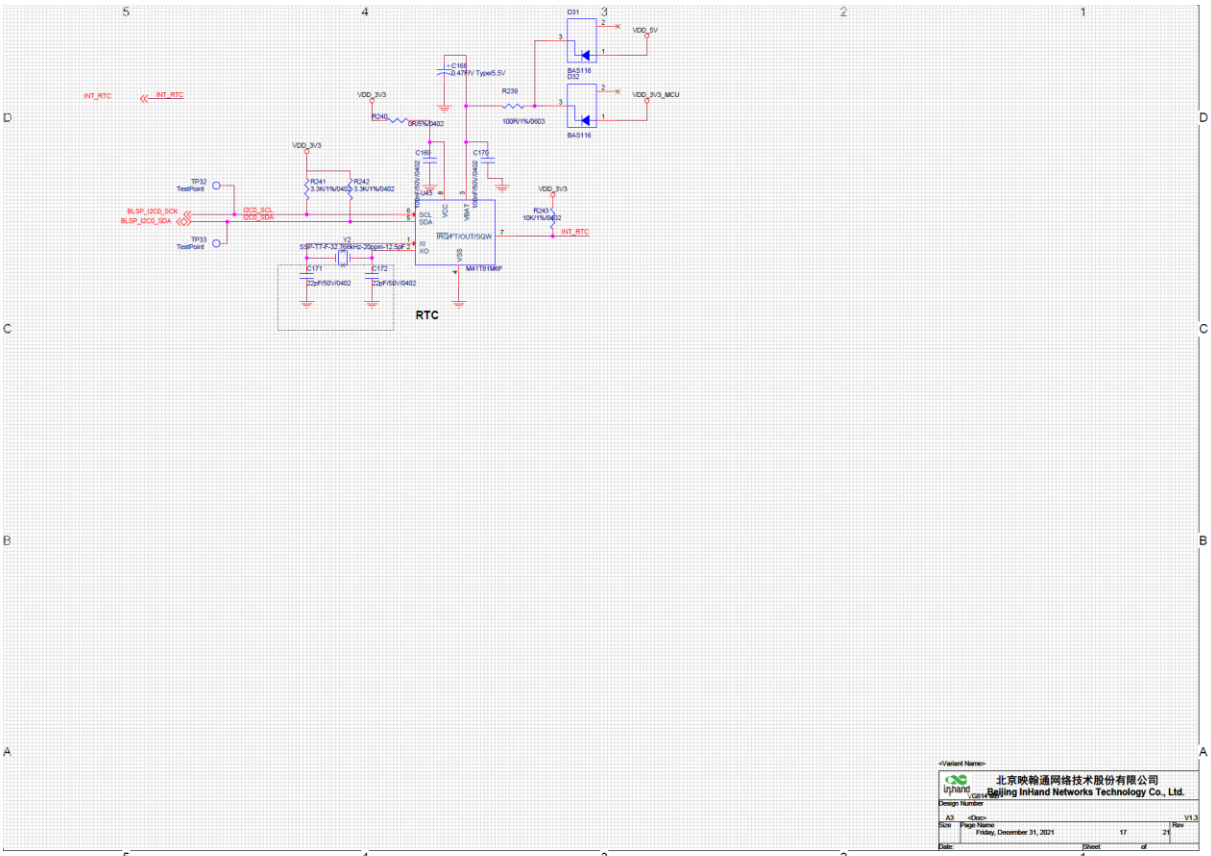

MAX: 4.5A, M2_VCC
USE 3.8V
0.6*(453/100+1)=3.318V
0.6*(536/100+1)=3.816V
MODULE2
MODULE2 SIM CARD
SW:SIM2
SW:SIM1
Logic=0 : COM=NC
Logic=1 : COM=NO
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.
Friday, December 31, 2021

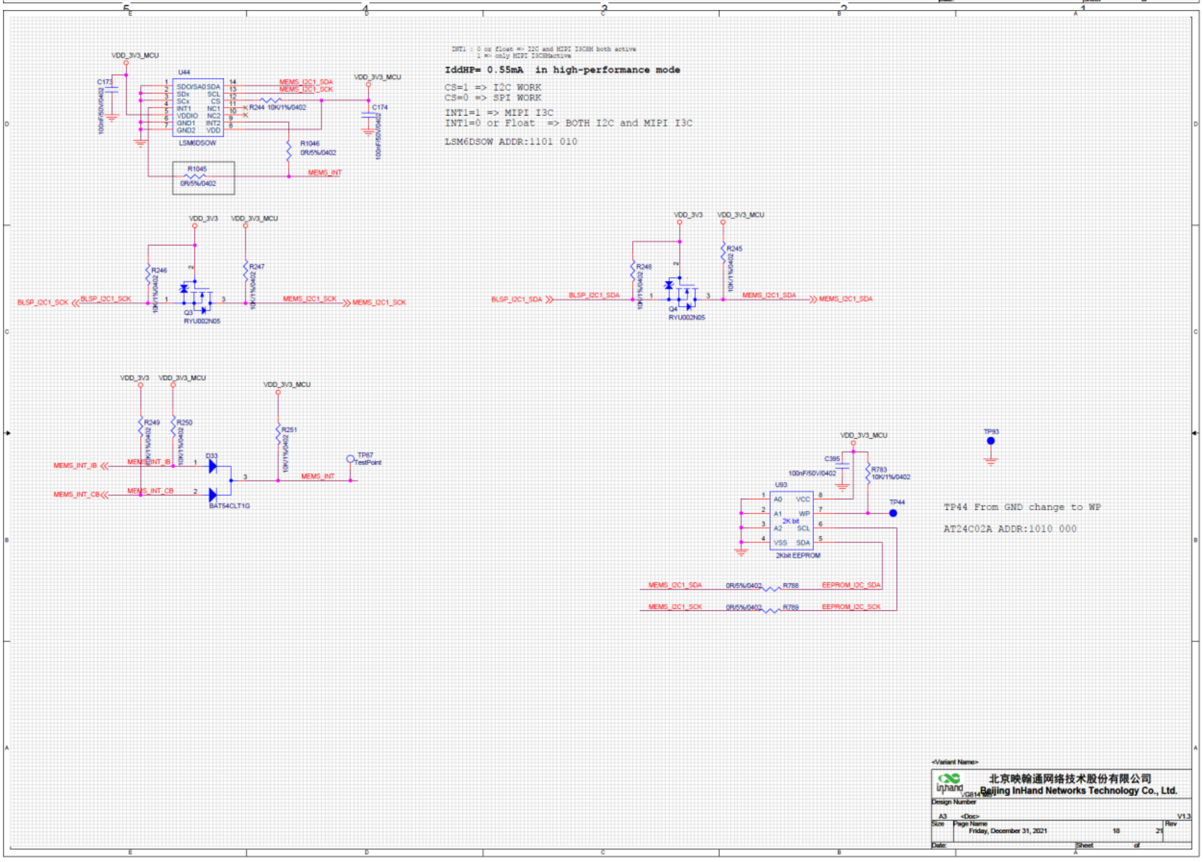





CAN1
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.


SYSTEM
Wifi-2.4G
WWAN1 Cellular
Wifi-5G
WWAN1 Signal
WWAN2 Cellular
GNSS
WWAN2 Signal
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.


RTC
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.


IddHP= 0.55mA in high-performance mode
CS=1 => I2C WORK
CS=0 => SPI WORK
INT1=1 => MIPI I3C
INT1=0 or Float => BOTH I2C and MIPI I3C
LSM6DSOW ADDR:1101 010
TP44 From GND change to WP
AT24C02A ADDR:1010 000
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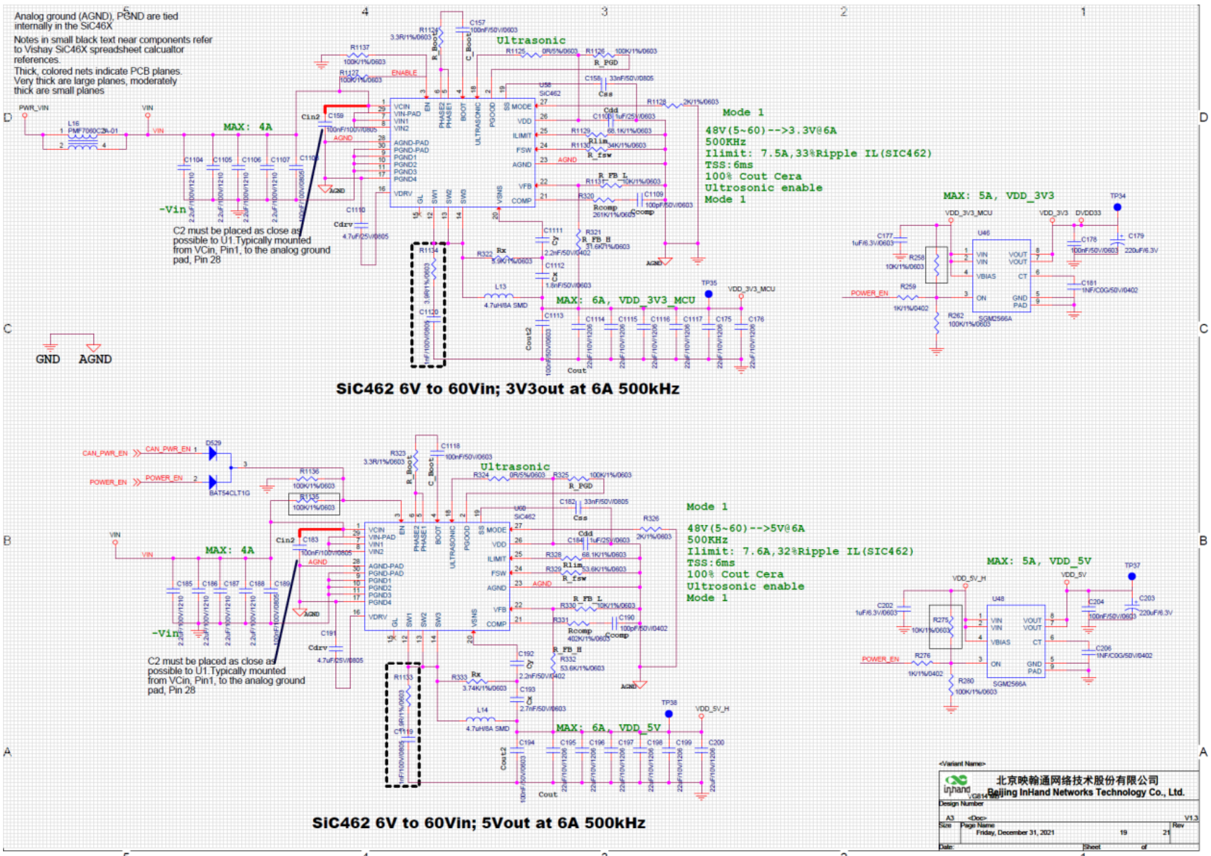
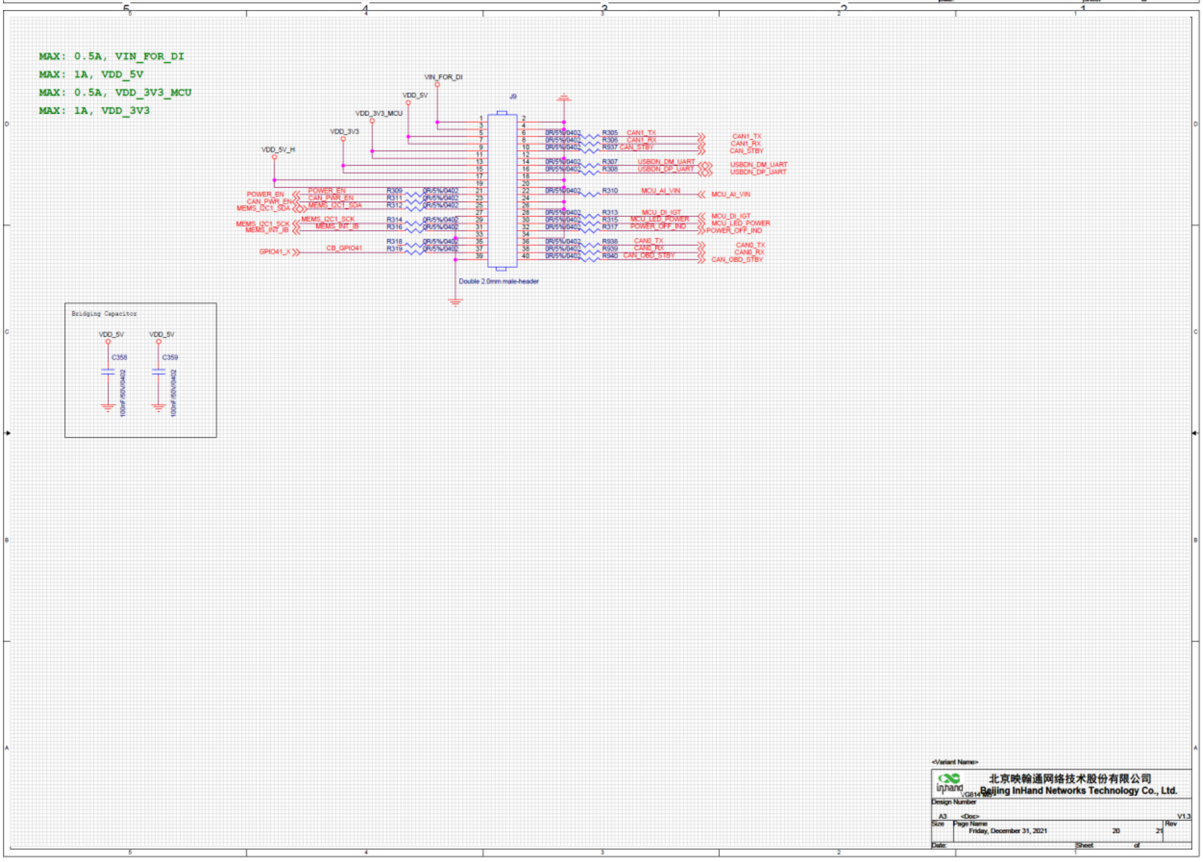
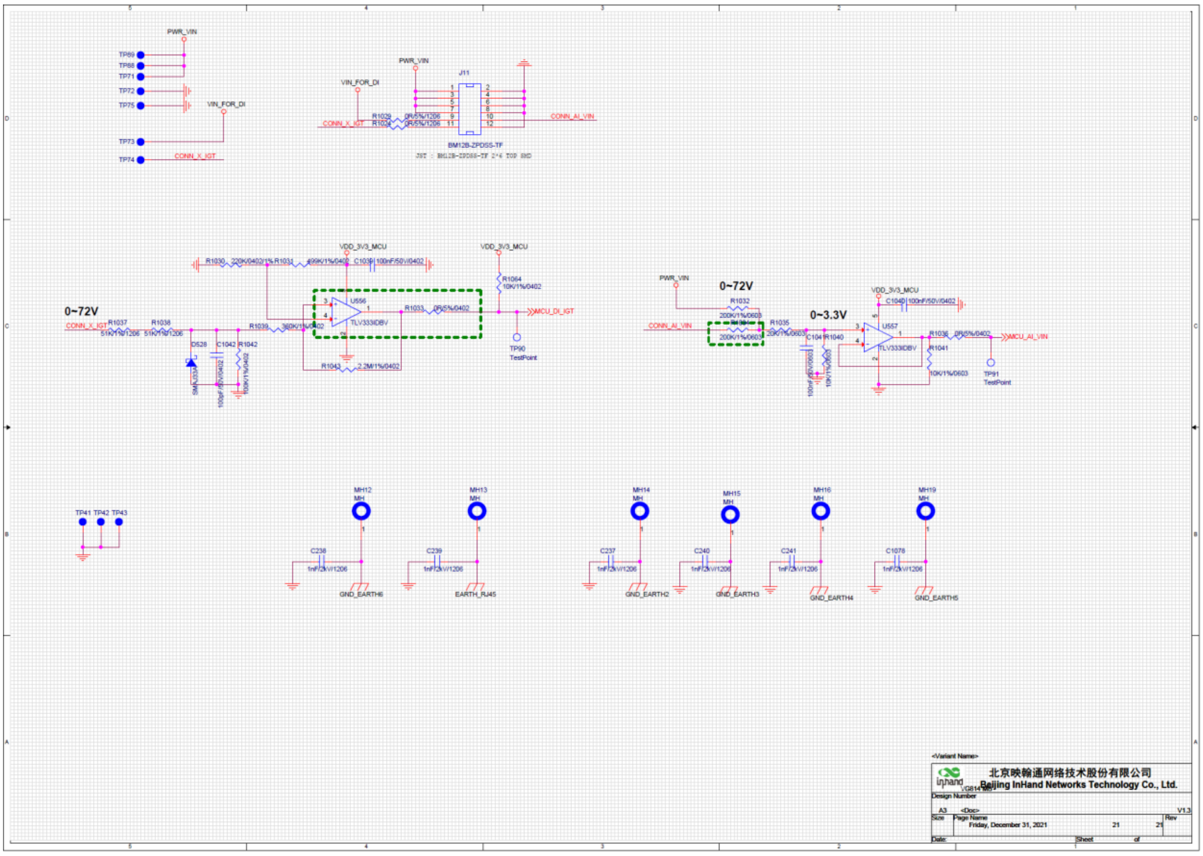
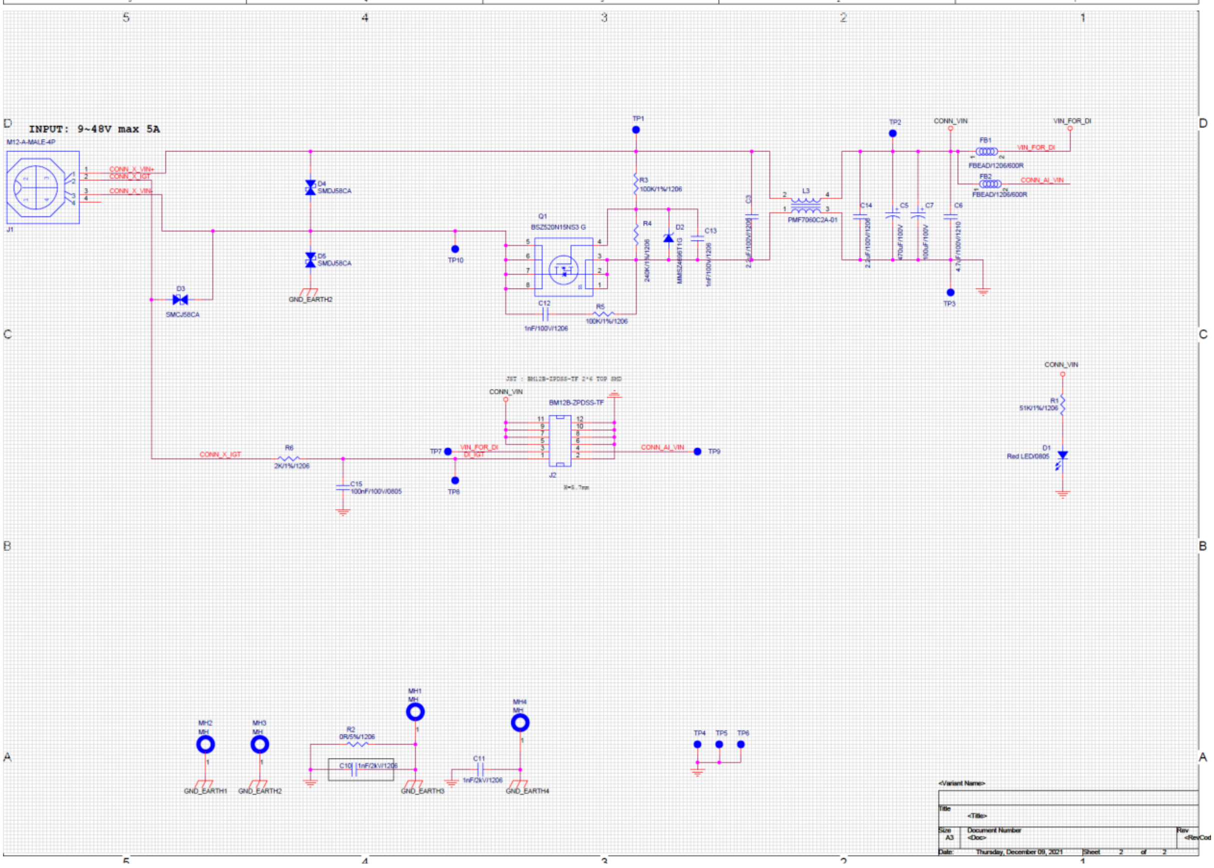

Analog ground (AGND), PGND are tied internally in the SiC46X
Notes in small black text near components refer to Vishay SiC46X spreadsheet calculator references.
Thick, colored nets indicate PCB planes. Very thick are large planes, moderately thick are small planes
MAX: 4A
Ultrasonic
Mode 1
48V(5~60)-->3.3V@6A
500KHz
Ilimit: 7.5A,33%Ripple IL(SIC462)
TSS:6ms
100% Cout Cera
Ultrosonic enable
Mode 1
MAX: 5A, VDD_3V3
MAX: 6A, VDD_3V3_MCU
C2 must be placed as close as possible to U1.Typically mounted from VCin, Pin1, to the analog ground pad, Pin 28
GND
AGND
SiC462 6V to 60Vin; 3V3out at 6A 500kHz
MAX: 4A
Ultrasonic
Mode 1
48V(5~60)-->5V@6A
500KHz
Ilimit: 7.6A,32%Ripple IL(SIC462)
TSS:6ms
100% Cout Cera
Ultrosonic enable
Mode 1
MAX: 5A, VDD_5V
MAX: 6A, VDD_5V
C2 must be placed as close as possible to U1.Typically mounted from VCin, Pin1, to the analog ground pad, Pin 28
SiC462 6V to 60Vin; 5Vout at 6A 500kHz
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.


MAX: 0.5A, VIN_FOR_DI
MAX: 1A, VDD_5V
MAX: 0.5A, VDD_3V3_MCU
MAX: 1A, VDD_3V3
Double 2.0mm male-header
Bridging Capacitor
北京映翰通网络技术股份有限公司
Beijing InHand Networks Technology Co., Ltd.





List of components constituting the ESA

Part_Number	Part Reference	PCB Footprint	Value	Quantity
EPCB240026			PCB	1
ECAP050063	C1 C2 C57 C64 C65 C66 C67 C68 C235	C0402	10pF/50V/0402	9
ECAP050150	C3 C4 C5 C6 C11 C12 C13 C14 C15 C16 C19 C20 C21 C22 C24 C25 C26 C27 C28 C29 C30 C31 C33 C34 C35 C36 C37 C38 C39 C40 C42 C44 C49 C52 C58 C97 C100 C105 C113 C115 C120 C155 C162 C163 C164 C165 C166 C167 C169 C170 C173 C174 C225 C226 C227 C336 C349 C350 C351 C352 C353 C354 C355 C356 C357 C358 C359 C395 C467 C468 C471 C472 C473 C474 C475 C477 C479 C1039 C1040 C1043 C1044 C1061 C1062 C1063 C1064 C1065 C1079 C1092 C1093 C1132 C1133	C0402	100nF/50V/0402	91
ECAP050002	C7 C8 C9 C10 C47 C237 C238 C239 C240 C241 C341 C1078	C1206	1nF/2kV/1206	12
ECAP050193	C17 C18 C111 C112 C114 C116 C118 C119	C0402	33pF/50V/0402	9
ECAP050099	C32 C53 C177 C202 C463 C476	C0603	1uF/6.3V/0603	6
ECAP050144	C41	C1210	100uF/10V/1210	1
ECAP050177	C43 C46 C48 C51 C55 C99 C103 C104 C175 C176 C195 C196 C197 C198 C199 C200 C1114 C1115 C1116 C1117	C1206	22uF/10V/1206	20
ECAP050134	C45 C160 C161 C338 C339	C0402	22pF/50V/0402	5
ECAP050176	C50 C59 C60 C71 C94	C0805	10uF/10V/0805	5
ECAP050009	C54 C157 C178 C194 C204 C1041 C1113 C1118	c0603	100nF/50V/0603	8
ECAP050148	C56 C181 C206	C0402	1NF/C0G/50V/0402	3
ECAP050064	C61 C63 C190 C1042 C1076 C1077	C0402	100pF/50V/0402	6
ECAP050149	C62	C0402	47pF/50V/0402	1
ECAP020014	C98 C179 C203	STC7343	220uF/6.3V	3
ECAP050235	C101	C0402	3.3nF/50V/0402	1
ECAP050093	C102	C0402	15pF/50V/0402	1
ECAP050004	C110 C117	c0603	220nf/25V/0603	2
ECAP050204	C158 C182	C0805	33nF/50V/0805	2
ECAP050123	C159 C183 C189 C1108	C0805	100nF/100V/0805	4
ECAP060009	C168	SCAP-V	0.47F/V Type/5.5V	1
ECAP050010	C184 C1103	C0603	1uF/25V/0603	2
ECAP050084	C185 C186 C187 C188 C1104 C1105 C1106	C1210	2.2uF/100V/1210	8
ECAP050060	C191 C469 C1110	C0805	4.7uF/25V/0805	3
ECAP050106	C192 C1111	C0402	2.2nF/50V/0402	2
ECAP050007	C193	c0603	2.7nF/50V/0603	1
ECAP050209	C233	C0402	1uF/25V/0402	1
ECAP050242	C343	C0603	100nF/100V/0603	1
ECAP050129	C464 C466	C0402	18pF/50V/0402	2
ECAP050065	C470 C478	C0402	10nF/50V/0402	2
ECAP050138	C1089 C1090 C1091	C0603	10uF/10V/0603	3
ECAP050025	C1109	c0603	100pF/50V/0603	1
ECAP050130	C1112	C0603	1.8nF/50V/0603	1
ECAP050035	C1119 C1120	C0805	1nF/100V/0805	2
EDIO010066	D1 D2 D3 D4	LED1206_RIGHTANGLE	Yellow LED/1204/side	4
ESEP000063	D6 D26	SODFLN90X60X55L25X50N	UAD8C05L01	2
EDIO010008	D8 D46	LED0603	Green LED/0603	2
EDIO000027	D11	sot23	BAT54A	1
EDIO000019	D12 D33 D529	SOT23	BAT54CLT1G	3
EDIO000020	D13	SOD323_AC	BAT54HT1G	1
EDIO040001	D14 D15 D18 D19 D22 D23 D27 D28 D32	SOT23	BAS116	9
EDIO010092	D16 D17 D20 D21 D24 D25 D29 D30	LED4-3_1X1_6-H1_1	LED,RGB	8
ESEP000079	D528	SMA_DO214AC_C2A1	SMAJ33A	1
EIND050001	FB1	10603	Bead/0603	1
EIND050004	FB2	10805	Bead/0805	1
EIND050020	FB6 R102	2P0805R	600R@100MHz/2A/0805	2
ECON010033	J1	61082_201	61082-201402PLF	1
ECON070063	J4 J5 J6 J7	CON9-BST_J455_38_01	M12-X-F-8P	4
ECON050123	J9	CON-2X20-2P0	Double 2.0mm male-header	1
ECON040070	J11	BM12B-ZPDSS-TF	BM12B-ZPDSS-TF	1
ECON090060	J13 J14 J15 J16 J17	P1201Y	P1201Y	5
EIND040012	L1 L2 L3 L7	ind4_choke_0805	900HM 100MHz 220MA	4
EIND000059	L4	1_ind_smd_v1s252012e	2.2uH/2A	1
EIND080001	L5	L0402	27nH/0402	1
EIND080005	L6	L0402	6.8nH/5%/0402	1
EIND000075	L9	INDM730X680X300L160X300N	4.7uH/5.5A	1
EIND000090	L13 L14	L-SPM1040	4.7uH/8A SMD	2
EIND040006	L16	PMF7060	PMF7060C2A-01	1
APSS010101	MH1 MH2 MH3 MH4	MTGP620H422	SMTSOM3-5.5	4

APSS010076	MH7 MH8	MTGP620H373	SMTSOM2-3	2
ETSR000007	Q2 Q9	SOT95P280X110-3	DTC144EKA	2
EFET010013	Q3 Q4	SOT323_3	RYU002N05	2
EFET010002	Q8	SOT23	2N7002	1
ERES040213	R1 R2 R3 R4 R5 R6 R7 R8 R9 R10 R11 R12 R13 R14 R15 R16 R17 R18 R19 R20 R21 R22 R23 R24 R25 R26 R27 R28 R29 R30 R31 R32 R33 R34 R35 R36 R37 R38 R39 R40 R41 R42 R43 R44 R45 R46 R47 R48 R49 R50 R51 R55 R57 R58 R59 R60 R61 R63 R64 R86 R87 R88 R89 R94 R95 R114 R139 R146 R151 R153 R165 R240 R305 R306 R307 R308 R309 R310 R311 R312 R313 R314 R315 R316 R317 R318 R319 R479 R481 R482 R483 R484 R485 R788 R789 R937 R938 R939 R940 R973 R978 R979 R1004 R1005 R1006 R1007 R1027 R1028 R1033 R1036 R1046	r0402	0R/5%/0402	111
ERES040101	R52 R54 R96 R326 R1128	R0603	2K/1%/0603	5
ERES040130	R53 R56 R62 R106 R107 R108 R239	R0603	100R/1%/0603	7
ERES040089	R65 R80 R330 R1040 R1041 R1131	R0603	10K/1%/0603	6
ERES040038	R66 R67 R68 R69	R0603	75R/5%/0603	4
ERES040271	R70 R71 R72 R73 R134 R159 R209 R211 R213 R215 R216 R218 R220 R222 R224 R226 R227 R229 R230 R231 R232 R233 R234 R235 R236 R237 R238 R259 R276 R494 R1062 R1069	R0402	1K/1%/0402	32
ERES040044	R74 R1048	R0603	12K/1%/0603	2
ERES040341	R77 R81	R0402	200K/1%/0402	2
ERES040283	R83 R84 R478 R1042 R1047 R1123	R0402	100K/1%/0402	6
ERES040276	R90 R121 R130 R132 R140 R147 R154 R164 R195 R196 R198 R200 R201 R202 R203 R205 R206 R207 R208 R210 R225 R243 R244 R245 R246 R247 R248 R249 R250 R251 R400 R405 R406 R407 R408 R409 R410 R488 R489 R496 R497 R498 R783 R976 R1049 R1064 R1067 R1070 R1071 R1073 R1074 R1092 R1093 R1094 R1142	R0402	10K/1%/0402	55
ERES040154	R91	R0603	110K/1%/0603	1
ERES040088	R93 R101 R262 R280 R325 R1063 R1126 R1127 R1136 R1137	R0603	100K/1%/0603	10
ERES040330	R99 R1035	R0603	20K/1%/0603	2
ERES040078	R103	R0603	10R/1%/0603	1
ERES040360	R109 R110	R0402	27R/1%/0402	2
ERES040240	R113 R144 R192 R197 R228 R401 R402	R0402	33R/1%/0402	7
ERES040518	R148	R0402	1. 3R/5%/0402	1
ERES040347	R149 R329 R332	R0603	53. 6K/1%/0603	3
ERES040390	R241 R242	RESC1005_IS0402Q	3. 3K/1%/0402	2
ERES040036	R320	R0603	261K/1%/0603	1
ERES040251	R321	R0603	31. 6K/1%/0603	1
ERES040024	R322	R0603	5. 9K/1%/0603	1
ERES040328	R323 R1124	R0603	3. 3R/1%/0603	2
ERES040001	R324 R1125	R0603	0R/5%/0603	2
ERES040152	R328 R1129	R0603	68. 1K/1%/0603	2
ERES040321	R331	R0603	402K/1%/0603	1
ERES040257	R333	R0603	3. 74K/1%/0603	1
ERES040034	R1024 R1029	R1206	0R/5%/1206	2
ERES040528	R1030	R0402	220K/0402/1%	1
ERES040316	R1031	R0402	499K/1%/0402	1
ERES040033	R1032	R0603	200K/1%/0603	1
ERES040555	R1037 R1038	R1206	51K/1%/1206	2
ERES040468	R1039	R0402	360K/1%/0402	1
ERES040272	R1043	R0402	2. 2M/1%/0402	1
ERES040229	R1122	R1206	100R/5%/1206	1
ERES040150	R1130	R0603	34K/1%/0603	1
ERES040138	R1133 R1134	R0603	3. 9R/1%/0603	2
ESWT000005	S1	switch_4legs	reset button(outward)	1
ESEP000065	U1 U2 U5 U6 U7 U8 U9 U10 U12 U13 U31 U32	SON10P50_245X100X65L36X20	UBQ10A05L04HI	12
ETFR050012	U3 U4	DIP480W40P178L3250H1160Q36	LG2P109NM	2
EICP020109	U11	VQFN_57_700x700x100_40P	USB5744I	1
EICP030149	U14	SOT23-5	SY8089A	1
EICP030120	U15	SOT23_6L	AP2553W6-7	1
EICP030225	U16 U46 U48	SON9P50_200X200X80L30X25T160X90	SGM2566A	3
EMOD000050	U17	24-pin_stamp_holes	NEO-M8U	1
ECON060030	U18	UFLR-MINIPCI	RF antenna connector	1
EICP070016	U19	SMD5	SAFFB1G56KB0F5F, SAFFB1G56KB0F0A	1

EICP070031	U20	dfn50p150x100x80-6	AT2659	1
ECON090091	U27	minipcie_new_noMH	minipcie socket 5.2mm/2.9mm	1
EICP030222	U28	SOT23-6	SY8113IADC	1
EICP040080	U29	WQFN16_3X3_0_5P	FSA2567MPX	1
EICP020133	U37 U54	soic127p1155x750x265-18	MCP2515T-I/SO	2
EXTL010033	U38 U56	CRYSTAL32X25	20MHz	2
EICP020047	U39 U40 U41	TSSOP-16	PCA9554	3
EICP040069	U42	tssop14	SN74HC164PWR	1
EICP050006	U43	SOIC_8	M41T81M6F	1
EICP080002	U44	lga50p250x300x86-141	LSM6DSOW	1
EICP030198	U58 U60	vishay_mlp55-271	SiC462	2
EICP020080	U77	QFN_6X6_36	USB2514Bi-AEZG	1
EICP010042	U93	SOIC8	2Kbit EEPROM	1
EICP060037	U556 U557	SOT23-5	TLV333IDBV	2
ECON090092	U559 U560	SIM-MUP-C720-1	MUP-C720-1, PUSH-PUSH	2
ECON100027	USB1	USB3-AF90	USB3.0	1
EXTL010021	Y1	CRYSTAL5X3_4	25M/20pF/20ppm	1
EXTL010031	Y2	SSP-T7-F	SSP-T7-F-32.768kHz-20ppm-12.5pF	1
EXTL010024	Y5	CRYSTAL32X25	XTAL/3.2x2.5/24MHz	1